

Competence for transcellular infection in the root cortex involves a post-replicative, cell-cycle exit decision in *Medicago truncatula*

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Abstract

During root nodule symbiosis (RNS), cell-division activity is re-initiated and sustained in the root cortex to create a hospitable cellular niche. Such temporary and spatially confined site is required to render host cells compatible with the intracellular progression of rhizobia. Although it has been suggested that early infection events might involve a pre-mitotic cell-cycle arrest, this process has not been dissected with cellular resolution. Here, we show that a dual-colour *Medicago* histone reporter robustly identifies cells with different mitotic or endoreduplication activities in the root cortex. By imaging deep root tissues, we found that a confined trajectory of cortical cells that are transcellularly passed by infection threads are in a stage of the cell-cycle that is distinct from directly adjacent cells. Distinctive features of infected cells include nuclear widening and large-scale chromatin rearrangements consistent with a cell-cycle exit prior to differentiation. Using a combination of fluorescent reporters demarcating cell-cycle phase progression, we confirmed that a reduced proliferation potential and modulating the G2/M transition, a process possibly controlled by the NF-YA1 transcription factor, mark the success of rhizobial delivery to nodule cells.

eLife assessment

This is a **fundamental** cell biological study of host responses during symbiotic microbial infection of plants. **Compelling** imaging-based approaches using genetically-encoded cell cycle markers show that in *Medicago truncatula* root cortex cells, early rhizobial infection events are associated with cell-cycle re-entry, but once the infection is established, host cells exit the cell cycle. The work will be of interest to a wide range of colleagues, from development and cell biology to plant-microbe interactions.

Introduction

Cell-cycle regulation is a key pathway to maintain beneficial and sustainable symbiotic associations that are of primary importance in both marine and terrestrial ecosystems. One common theme to stabilize facultative associations is to achieve a mutual control of cell proliferation and differentiation dynamics in both the host and the symbionts (Kondorosi & Kondorosi, 2004 [↗](#); Russo & Genre, 2021 [↗](#)). Prime examples are flowering plants from a single phylogenetic clade, comprising four orders (Fabales, Fagales, Cucurbitales, Rosales) with legumes being the most prominent ones, that evolved the unique ability to accommodate soil-borne rhizobia intracellularly (Parniske, 2018 [↗](#)). Within this symbiosis, rhizobia fix atmospheric nitrogen and deliver ammonium to the host in exchange for plant photosynthates. The conversion of atmospheric dinitrogen into a usable form is highly energy-demanding and requires a low-oxygen environment, which is provided within nodules, specialized lateral organs formed on the roots that are densely populated by rhizobia. A widely adopted mechanism to access the developing nodule cells is the formation of cortical infection threads, transcellular conduits formed upon the invagination of the host plasma membrane and hijacking the cellular machinery of infected root cortical cells (Timmers et al., 1999 [↗](#); Su et al., 2023 [↗](#)). Although our molecular and cellular understanding of the infection mechanisms is expanding (Tsyganova et al., 2021 [↗](#)), our knowledge on how these processes are interconnected with the control of cell-cycle progression in cells hosting intracellular infection threads has remained fragmentary.

Plants engaging into arbuscular mycorrhizal (AM) or root nodule symbioses (RNS) have recruited a common range of cell-cycle processes (Foucher & Kondorosi, 2000 [↗](#); Russo & Genre, 2021 [↗](#)). In *Medicago truncatula* (hereafter, *Medicago*), the perception of and colonization by the AM fungus result in occasional cell divisions in the inner cortical layers (Russo et al., 2019 [↗](#)) and in diffuse endoreduplication events as intraradical hyphae spread throughout the root cortex (Carotenuto et al., 2019a [↗](#); Carotenuto et al., 2019b [↗](#)). During RNS, the perception of rhizobia results in the onset of a signalling cascade that triggers an outward gradient of cell-cycle reactivation and cell divisions. Although this is initiated in the pericycle, cell-cycle reactivation occurs throughout the inner cortical (C) layers C5, C4 and the middle C3 layer in *Medicago* (Timmers et al., 1999 [↗](#); Xiao et al., 2014 [↗](#)). The successful establishment of RNS further involves the tight coordination of the infection and organogenesis programmes (Guan et al., 2013 [↗](#)). It requires the progression of cortical infection threads through the C3 layer prior to subsequent periclinal divisions that produce a persistent nodule meristem (Xiao et al., 2014 [↗](#)). The resulting indeterminate nodules will later form well-defined developmental zones, with actively dividing meristematic cells giving rise to post-meristematic, differentiating central tissues of increasing ploidy (Vinardell et al., 2003 [↗](#); Nagymihály et al., 2017 [↗](#)). While those cells can be intracellularly colonized, the infection competence of nodule-like structures as being induced by exogenous application of cytokinin (Gauthier-Coles et al., 2019), constitutive activation of common symbiotic components (Gleason et al., 2006 [↗](#); Singh et al., 2014 [↗](#)) or the expression of meristematic factors (Dong et al., 2021 [↗](#)) may be massively reduced in many of these spontaneous nodules (Hayashi et al., 2010 [↗](#); Liu et al., 2022 [↗](#)). The lack of infectibility cannot be overcome even when applying bacterial strains which enter plant cells from the intercellular space (Liang et al., 2019 [↗](#)). This clearly demonstrates the ultimate requirement of cells being in a so far uncharacterized state of infection competence.

Transcriptome profiling of *Medicago* root hairs, treated with rhizobial Nod factors or undergoing rhizobial colonization, uncovered changes in several cell-cycle related genes (Breakspear et al., 2014 [↗](#); Liu et al., 2019a [↗](#)). This perception of Nod factors results in the increased transcription of D-type cyclins, genes known to respond to extrinsic signals and to promote cell-cycle progression from the first GAP (G1) phase to the DNA-synthesis (S) phase (Dewitte et al., 2003 [↗](#)). Subsequently, the initiation of infection threads coincides with the highest expression of genes required for DNA replication, modification and repair, strongly suggesting that the epidermal infection programme

involves cell cycle re-entry and progression to a post-replicative phase (Breakspear et al., 2014). A repression of the endocycle was further proposed to occur since infected root hairs accumulate transcripts of the *OSD1/UVI4* gene (Breakspear et al., 2014), whose activity prevents an unscheduled increase in ploidy by inhibiting the anaphase-promoting complex/cyclosome (Iwata et al., 2011). However, much less is known about the transcriptional signature of infected cells along the cortical trajectory of transcellularly progressing infection threads. Such ability to guide rhizobia via cortical infection threads is yet thought to be an important innovation since it has been evolutionary maintained in the vast majority of nodulating species as it, most likely, maximizes the host control over bacterial delivery (Parniske, 2018; Cathebras et al., 2022 – pre-print).

A central regulator for controlling infection and organogenesis is the transcription factor *Nodule Inception* (*NIN*). Co-opted from nitrate response circuits, *NIN* acts as a master coordinator of infection, nodule organogenesis and nodule number (Marsh et al., 2007; Fournier et al., 2015; Soyano et al., 2014; Cathebras et al., 2022 – pre-print). It promotes the expression of several growth targets (Liu et al., 2019a) including *NF-YA1* (Soyano et al., 2013; Feng et al., 2021), a subunit of the conserved heterotrimeric Nuclear Factor-Y (NF-Y) transcription factor. A-, B- and C-type subunits of the NF-Y complex act as specialized modules, interacting with other transcription factors and recruiting chromatin remodelling enzymes to adjust cell specification to environmental or developmental needs (Zanetti et al., 2016; Myers & Holt, 2018). In vertebrates, the CCAAT-binding NF-Y transcription factor targets genes involved in all major activities executed in G2 and mitosis (M; Linhart et al., 2005) and is an important regulator of the G2/M transition (Manni et al., 2001). In legumes, the rhizobium-induced NF-YA1 subunit functions in cortical infection thread progression, in the establishment and maintenance of the nodule meristem and in the early differentiation of nodule cells (Comber et al., 2006; Laporte et al., 2014; Xiao et al., 2014; Hossain et al., 2016; Schiessl et al., 2023 – pre-print). Yet, whether the Medicago NF-YA1 protein regulates the G2/M transition is not known.

The hypothesis that intracellular infection at least partially relies on cell-cycle control is further supported by structural re-arrangements in cortical cells anticipating transcellular IT progression. These include the formation of pre-infection threads (PITs), which are transvacuolar, cytoplasmic bridges comparable to those occurring in pre-mitotic cells (van Brussel et al., 1992; Timmers et al., 1999). *In situ* detection of marker transcripts revealed that outer cortical cells recruit histone H4 used during DNA replication but do not express a B-type, mitotic cyclin gene – implying that activated target cells stop in the G2 phase of the cell cycle (Yang et al., 1994). G2-phase cell-cycle arrests have been repeatedly described in eukaryotic cells as appropriate gates for executing cell fate and patterning decisions (Meserve & Duronio, 2017), for wound healing (Cosolo et al., 2019) or for host infection during pathogenic and biotrophic interactions (Wildermuth et al., 2017). In plants, transiently or sustainably reducing the potential to enter mitosis is an appropriate response when cells must, for instance, cope with genotoxic stresses (Chen et al., 2017; Takahashi et al., 2019) or switch to another cell-cycle program, such as the meiotic (Hernandez-Lagana et al., 2021) or the endoreduplication cycle (De Veylder et al., 2011; Wildermuth et al., 2017). Modulating the G2/M transition combines well-described transcriptional and post-translational mechanisms, leading to the repression of genes necessary to execute cell division (Berckmans & De Veylder, 2009; Kobayashi et al., 2015), controlling the activity of cyclin-dependent kinases (CDK) and the stability of transcriptional repressors and mitotic inducers (Kondorosi & Kondorosi, 2004; Chen et al., 2017). Although previous reports have suggested such cell-cycle patterns to occur in roots during rhizobial infections (Yang et al., 1994; Breakspear et al., 2014), this has never been spatially resolved at cellular level.

Here, we employed a combination of cell-cycle reporters to resolve functional steps of cell-cycle progression on the cortical infection thread trajectory with cellular and subcellular resolution. We demonstrate that the first generation of Medicago cells successfully internalizing rhizobia shows unique features, including a marked decrease in their histone H3.1 content and a reduced

competence for both cell division and chromosome segregation. We also show that tetraploid (4n) cells reaching an 8C DNA content are optimal for infection by rhizobia. Our results strongly suggest that, upon transcellular infection thread passage, cortical cells undergo a last GAP phase after DNA replication and exit to differentiation. Furthermore, we present first evidence that the symbiotically-induced NF-YA1 subunit, by controlling mitotic entry in a heterologous system, holds potential to contribute to this process.

Results

A reduced proliferative potential typifies cells supporting cortical infection thread progression

In *Arabidopsis thaliana* (hereafter, *Arabidopsis*), the incorporation and eviction dynamics of histone H3 variants have been used to identify cells with different cell division and reprogramming potentials, during organ patterning (Otero et al., 2016) or when acquiring reproductive competence (Hernandez-Lagana & Autran, 2020). To test whether gaining a stage of intracellular infection competence involves a similar modification of the cell-division potential and associated chromatin reorganization, we searched for Medicago genes coding for the replicative histone variant H3.1 and the replacement variant H3.3 (Figure 1—figure supplement 1A; Probst et al., 2020). The encoded Medicago H3.1 protein is identical to the one present in *Lotus japonicus* and *Hordeum vulgare* and, as in other flowering plants, differs from H3.3 by only four residues (Figure 1—figure supplement 1B; Shi et al., 2011). To check that these Medicago *H3* genes encode for *bona fide* H3.1 and H3.3 counterparts, native promoter-genomic fusions with fluorescent tags were expressed in *Agrobacterium rhizogenes*-induced transgenic roots, formed on composite plants that were inoculated with the compatible symbiont *Sinorhizobium meliloti* (each composite plant representing an independent transformant; Boisson-Dernier et al., 2001). We adopted a fixation, clearing and cell-wall counter-staining procedure (Ursache et al., 2018) that enabled us to perform detailed imaging deep inside the multi-layered root cortical tissues. Medicago H3 fusion proteins fully recapitulated the patterns observed in *Arabidopsis* roots (i.e., patchy pattern of H3.1 and constitutive presence of H3.3; Figure 1—figure supplement 1C–E) and showed the expected localization in euchromatin (diffuse labelling) and heterochromatic regions (subnuclear foci; Figure 1—figure supplement 1C–E; Ingouff et al., 2010; Shi et al., 2011; Otero et al., 2016). A low H3.1 content, typical for cells with a reduced proliferative potential, also enabled us to identify putative quiescent centre (QC) cells within the open root apical meristem of Medicago (Figure 1—figure supplement 1E; Rost, 2011; Otero et al., 2016; Xiao et al., 2019). Preferentially incorporated during DNA replication, the canonical histone H3.1 variant controls the maintenance of genome integrity (Davarinejad et al., 2022) and epigenetic inheritance (Jiang & Berger, 2017) and is consequently maintained at high levels in cells engaged in recurrent cell division or endoreduplication cycles (Figure 1A). Once cells exit the cell cycle, H3.1 is evicted and likely replaced by H3.3 to mediate cellular differentiation in various *Arabidopsis* cell types (Otero et al., 2016). In Medicago, increased levels of *H3.1* transcripts were coherently detected at an early stage of nodule primordia development (Figure 1—figure supplement 1F; Schiessl et al., 2019) and mRNAs accumulated only in meristematic (ZI) and infection zones of mature nodules (ZII_d and ZII_p; Roux et al., 2014), while *H3.3* transcription stayed high in the differentiation zone (ZIII; Figure 1—figure supplement 1G). We verified a corresponding differential accumulation of H3.1 and H3.3 tagged proteins at cellular resolution in nodule sections (Figure 1B). The double histone reporter proved also useful to distinguish actively endoreduplicating cells prior to fungal colonization (Carotenuto et al., 2019a; Carotenuto et al., 2019b) from fully differentiated, arbuscule-containing cells (Figure 1C). Since the selected Medicago histone variants recapitulated the expected transcriptional and localization patterns in well-defined developmental zones, we considered them as functional reporters of H3.1 and H3.3. In addition, the H3.1/H3.3 balance enables to track cortical cells re-entering the cell cycle and adapting their responses to both bacterial and fungal symbionts.

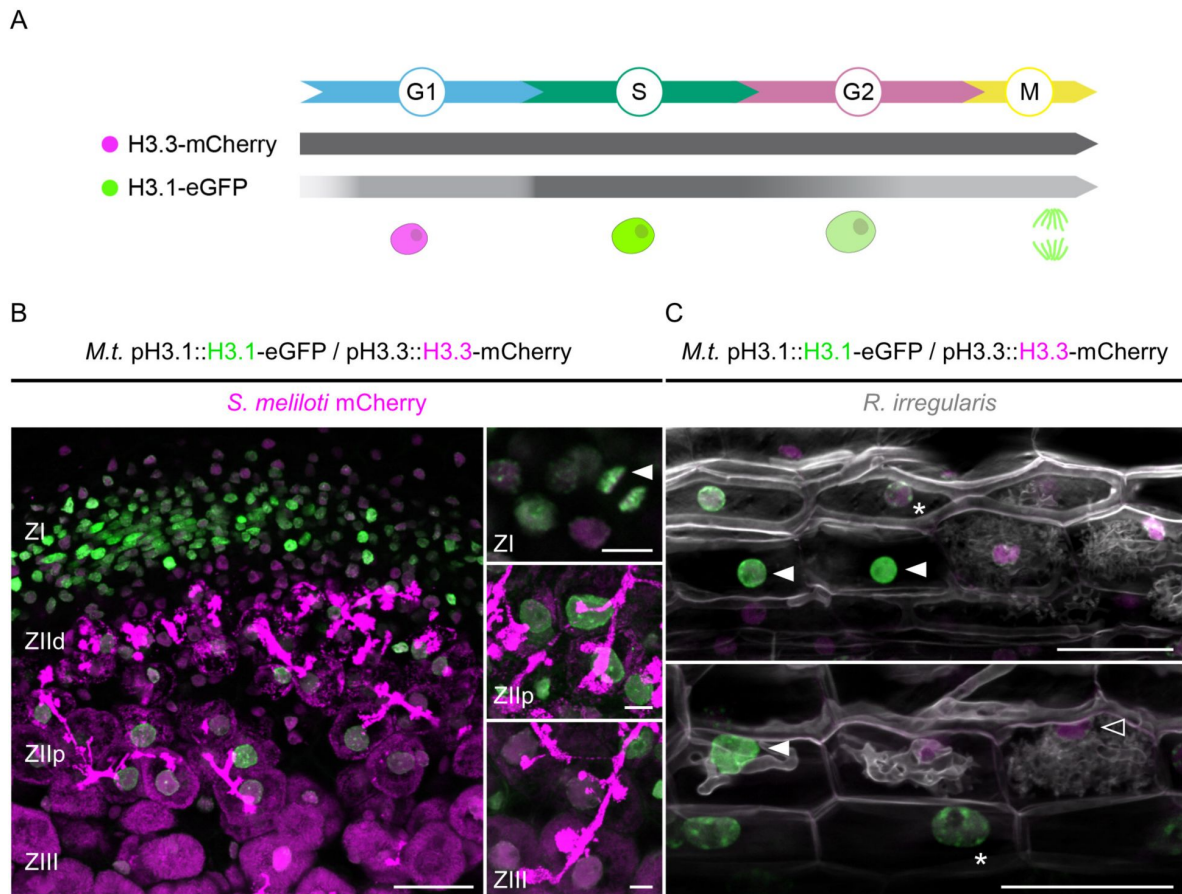


Figure 1.

Medicago histone H3.1 patterns reveal sustained mitotic and endocycling activities in a symbiotic context

(A) Schematic representation of histone H3.3-mCherry and H3.1-eGFP distribution (horizontal bars) and fluorescence intensity (grey saturation) throughout the different cell-cycle phases (adapted from Echevarria et al., 2021). H3.1 is predominantly expressed during S-phase and incorporated during DNA replication in proliferating (G1, S, G2, M) and endocycling (G1, S, G2) cells. H3.3 is constitutively produced. Nuclei of increasing size and DNA content are coloured according to their H3.3 (pink) and H3.1 (green) content in decondensed chromatin (G1, S, G2) or condensed chromosomes (M). (B) Confocal images of nodule sections isolated from WT transgenic roots inoculated with mCherry-producing *S. meliloti* (35 to 40 dpi), showing H3.1-eGFP (green) and H3.3-mCherry (magenta) localization across characteristic developmental zones. H3.1 accumulates in the meristematic zone (ZI) enriched in proliferating cells (late-anaphase chromosomes indicated by a filled arrowhead) and in regions with high endoreduplication activity (ZIId and ZIIp, distal and proximal parts of the infection zone) where rhizobia are released into membrane-bound compartments called symbiosomes. H3.1 is extensively replaced by H3.3 in the fixation zone (ZIII) where host cells and rhizobia complete their differentiation process. Images are maximum intensity projections except the top-right panel (single focal plane). Scale bars: left panel = 50 μ m; right panels = 10 μ m. Transformation experiments were repeated 3 times with a total of 9 nodules from 6 composite plants showing similar results. (C) Confocal images of whole-mount WT roots colonized by *Rhizophagus irregularis* (15 dpi). Plant and fungal cell walls were stained with Calcofluor white (grayscale). H3.1 was enriched in chromocenters (heterochromatin foci in nuclei indicated by stars) and kept at high levels in the euchromatin (diffuse labeling in nuclei pointed by filled arrowheads) from neighbouring (upper panel) and early-arbusculated cells (lower panel) of the inner cortical tissue. The empty arrowhead points to a nucleus with a low H3.1 content in a fully-arbusculated, differentiated cell. Images are maximum intensity projections. Scale bars: 50 μ m. Two independent transformation experiments were performed with 3 to 5 composite plants analyzed per replica.

We then focused on the H3.1 nuclear content at early interaction stages between rhizobia and the host plant (8 days post inoculation [dpi]; **Figure 2A–B**). In accordance with RNA sequencing data (**Figure 1**–figure supplement 1F; Schiessl et al., 2019), we observed the replicative H3.1 variant accumulating in regularly dividing cortical cells forming the nodule primordium (**Figure 2A**, left panel). The H3.1-eGFP fusion protein was also visible in cells penetrated but not fully passed by an infection thread (**Figure 2A**, close up). This is consistent with earlier observations that cells preparing for intracellular infection pass through S-phase (Yang et al., 1994). By contrast, host cells sustaining cortical infection thread progression in the middle and inner cortex (C3 and C4) up to the first recipient cells in the nodule primordium (C4/5-derived; **Figure 2B**) exhibited enlarged nuclei (**Figure 2C**) with a significantly decreased histone H3.1 content compared to their direct neighbouring cells (**Figure 2D**). Such pattern was also observed using another *H3.1* gene expressed in root tissues (*H3.1* (2); **Figure 2**–figure supplement 1A–E). Nuclear enlargement associated with a massive eviction of H3.1 typically identifies cells having completed a DNA replication round, but pausing in the following GAP phase for cell-cycle switches or exit decisions (**Figure 2E**; Otero et al., 2016; Hernandez-Lagana & Autran, 2020; Probst et al., 2020). Together, our data strongly suggest that cortical cells along the infection thread trajectory, different to their direct neighbours, exhibit a reduced proliferation potential and presumably exit the cell cycle (**Figure 2 B–E**). The observed nuclear enlargement and large-scale chromatin rearrangements are consistent with an increased transcriptional activity (Knaack et al., 2022) concomitant with sustained infection thread progression (Breakspear et al., 2014; Liu et al., 2019a). This proves the long-standing hypothesis that cortical cells competent for transcellular infection suspend their course in G2 (Yang et al., 1994; Timmers et al., 1999) in *Medicago*.

Exiting the cell cycle on the cortical infection thread trajectory aligns infection and organogenesis

Next, we tested whether a cell-cycle exit decision and associated remodelling of the histone H3 composition requires the tight coupling of infection and nodule organogenesis as observed under wild-type (WT) conditions. Thus, we compared the H3.1/H3.3 patterns in WT roots to those of two mutants in which nodule organogenesis is either abolished (*daphne-like*; Liu et al., 2019b) or where nodules form but cortical infection threads are impaired in their progression (*nf-ya1-1*; Laporte et al., 2014). In the WT situation, the majority of cortical cells (90%) enabling transcellular passage until reaching the nodule primordium showed a reduced proliferation potential (low H3.1 content, $n = 41$; **Figure 3A** and **Figure 3**–figure supplement 1A). A higher H3.1 content was occasionally observed in the last infected cell in the absence of nodule primordium organogenesis (NOD[−]; **Figure 3**–figure supplement 1B). Absence of inner cortical cell division is also a characteristic feature of the *daphne-like* *Medicago* mutant (**Figure 3**–figure supplement 1C), which lacks a remote cis-regulatory region in the *NIN* promoter perceiving cytokinin signals (Liu et al., 2019b). This non-nodulating mutant still produces WT-like root hair infections occasionally reaching outer cortical layers. In this context, the percentage of infected cortical cells visibly accumulating H3.1 in the euchromatin increased from 10% in the WT to almost 41% in the *daphne-like* mutant ($n = 44$; **Figure 3A** and **Figure 3**–figure supplement 1D). This indicates that outer cortical cells hosting an infection thread lack an extensive chromatin reorganization in the absence of sustained nodule organogenesis.

As a second mutant, we chose a *nf-ya1* null allele, producing smaller nodules that are additionally delayed and impaired in their development (Laporte et al., 2014; Xiao et al., 2014). Nodule primordia of the *nf-ya1-1* mutant showed reduced cell layers, with a lower frequency of cell divisions in the inner cortex (C4/5) and only few anticlinal divisions in the middle layer (C3; **Figure 3**–figure supplement 1E–G; Xiao et al., 2014; Schiessl et al., 2023 – pre-print). Cortical infection threads appeared abnormally bulbous (**Figure 3A** and **Figure 3**–figure supplement 1E–F) and showed signs of early abortion (**Figure 3**–figure supplement 1G; Laporte et al.,

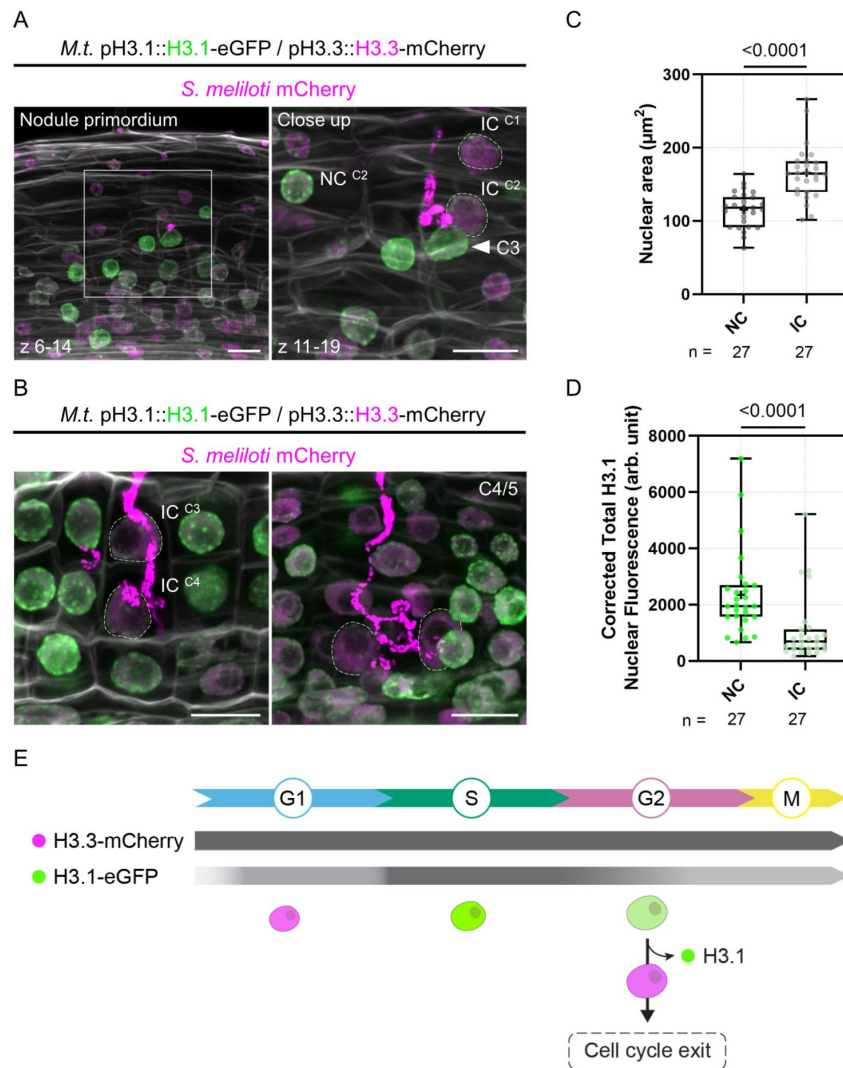


Figure 2.

Individual reprogramming for infection includes large-scale chromatin rearrangements

(A-B) Confocal images of whole-mount WT roots expressing the pH3.1::H3.1-eGFP / pH3.3::H3.3-mCherry reporter and inoculated with mCherry-producing *S. meliloti* (8 dpi). Images are maximum intensity projections (eGFP: green; mCherry: magenta; Calcofluor white cell-wall staining: grayscale). (A) Left panel: the solid-line frame indicates the outer and middle cortical regions of an early nodule primordium shown in the close up. Right panel: dotted-line regions of interest (ROIs) indicate nuclei of infected cells (IC) fully passed by a cortical infection thread in the first (C1) and second (C2) cortical layers. NC: neighbouring cell. The filled arrowhead points to the nucleus of a host cell from the middle cortical layer (C3) which is just penetrated by the infection thread. Scale bars: 20 μm . (B) Left panel: dotted-line ROIs indicate nuclei of infected cells (IC) being passed (C4) or recently passed (C3) by a cortical infection thread. Right panel: dotted-line ROIs indicate nuclei of infected cells in the inner cortex (C4/5) of a nodule primordium with several cell layers. Scale bars: 20 μm . (C-D) Quantification of the nuclear area (C) and corrected total H3.1-eGFP nuclear fluorescence (D) at the equatorial plane in couples of neighbouring (NC) and infected cells (IC) from the same cortical layer (C2 to C4, $n = 27$; see Materials and methods for more details). Roots from 2 to 7 composite plants with visible signs of inner cortical cell division (*i.e.*, showing high H3.1-eGFP signal) were analyzed from 3 independent transformation experiments. All data points are shown and crosses indicate sample means. Differences were statistically significant (p -values < 0.0001) using an unpaired t-test with Welch's correction (C) or a Mann-Whitney test (D). (E) Schematic representation of histone H3.3-mCherry and H3.1-eGFP distribution (horizontal bars) and fluorescence intensity (grey saturation) throughout the different cell-cycle phases. Nuclei of increasing size and DNA content are coloured according to their merged H3.3 (magenta) and H3.1 (green) content in decondensed chromatin (G1, S, G2). This diagram illustrates cells exiting the cell cycle within proliferating or endoreduplicating populations, which proceed to a massive H3.1 eviction following their last DNA replication round (Otero et al., 2016).

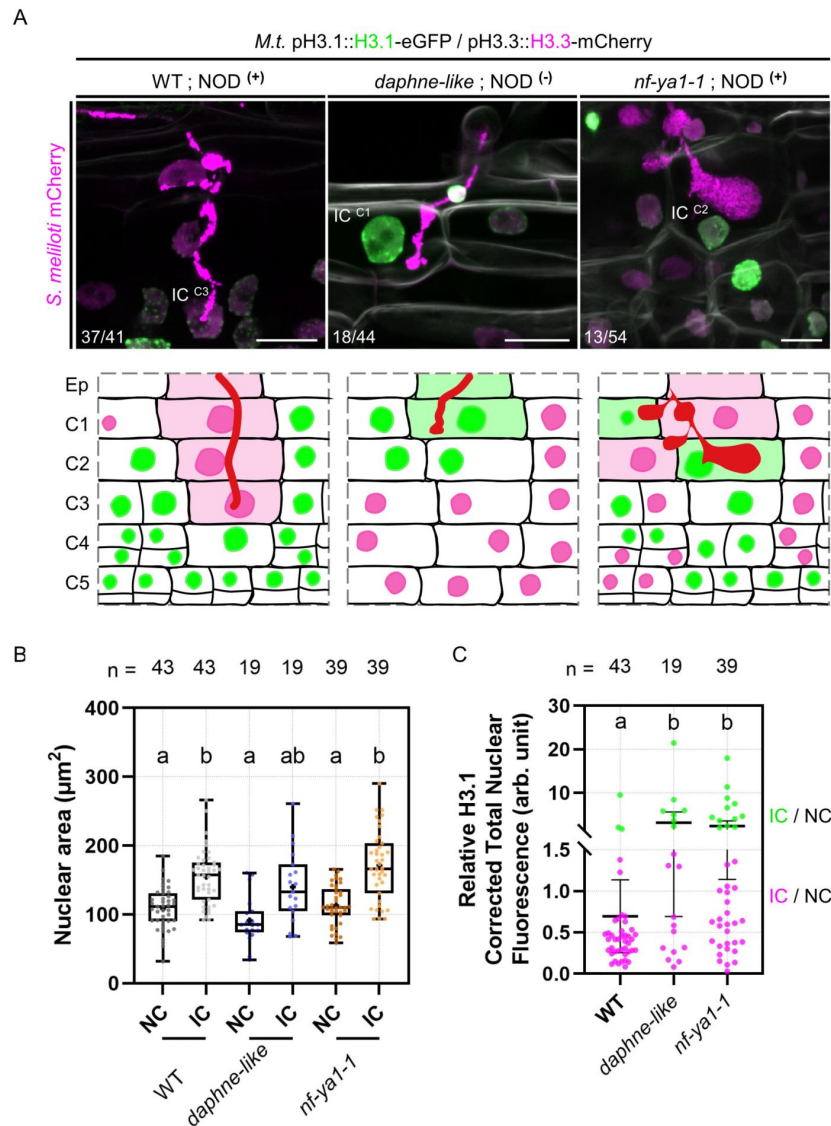


Figure 3.

H3.1 eviction coincides with cortical infection thread progression

(A) Confocal images of whole-mount transgenic roots co-expressing H3.1-eGFP and H3.3-mCherry in three different genotypes: WT (11 dpi), *daphne-like* (14 dpi) and *nf-ya1-1* (12 dpi), inoculated with mCherry-producing *S. meliloti*. WT and *nf-ya1-1* plants initiate nodule formation (NOD⁺) whereas the *daphne-like* mutant is non-nodulating (NOD⁻). Numbers indicate the frequencies of observation of a low (WT) or high (*daphne-like*, *nf-ya1-1*) H3.1-eGFP content in the last infected cortical cell. Corresponding schematic representations below each panel depict cellular proliferation or endocycling activities indicated by nuclear H3.1 levels (low: magenta; high: green) on the cortical infection thread trajectory (C1 to C3; diffuse magenta or green colouring) and in inner cortical layers (C4 to C5). Rhizobia inside infection threads are depicted in red. Confocal images are maximum intensity projections and show merged fluorescent channels (eGFP: green; mCherry: magenta; Calcofluor white cell-wall staining: grayscale in the middle and right panels). Scale bars: 20 µm. IC: infected cell. Ep: epidermis. **(B-C)** Quantification of the nuclear area **(B)** and the relative corrected total H3.1-eGFP nuclear fluorescence (IC/NC) **(C)** at the equatorial plane in couples of neighbouring (NC) and infected cells (IC) from the same cortical layer in WT (C1 to C4), *daphne-like* (C1 to C2) and *nf-ya1-1* (C1 to C4) genetic backgrounds (n = 43, 19 and 39 nuclei, respectively). Different letters indicate statistically significant differences according to a Kruskal-Wallis test followed by Dunn's multiple comparisons test. **(B)** All data points are shown and crosses indicate sample means. **(C)** All data points are shown with magenta or green indicating relative H3.1-eGFP nuclear signals (IC/NC) below 1.5 or above 1.5, respectively. Horizontal bars indicate sample means with 95% confidence interval. Ten to 21 composite plants (8-14 dpi) from 2 (*daphne-like*, *nf-ya1-1*) to 7 (WT) independent transformation experiments were analyzed.

2014 [\[1\]](#); Schiessl et al., 2023 [\[2\]](#) – pre-print). We observed that, even though the majority (76%) of crossed cortical cells were presumably arrested in a GAP phase (low H3.1 content, $n = 54$; **Figure 3** [\[3\]](#)–figure supplement 1F), such pausing was not observed in about 24% of the inspected events. Here, infection threads were either blocked or delayed in the outer cortex and the replicative H3.1 variant was retained (**Figure 3A** [\[4\]](#) and **Figure 3** [\[5\]](#)–figure supplement 1G). Although the last infected cortical cells showed a significant nuclear enlargement in all three genetic backgrounds investigated (**Figure 3B** [\[6\]](#)), failure to coordinate organogenesis and infection programs in the mutants resulted in an increased occurrence of infected cells accumulating more H3.1 than their direct neighbours (**Figure 3C** [\[7\]](#)). Together, these data reveal that an appropriate reduction in cell-cycle activity and associated chromatin changes in infected cells coincide with an efficient progression of cortical infection threads.

Cortical cells supporting transcellular infection control their commitment to cell division

In *Medicago* and other plants producing indeterminate nodules, a local control of mitotic activity could be especially important for rhizobia to timely cross the middle cortex, where sustained cellular proliferation gives rise to the nodule meristem (Xiao et al., 2014 [\[8\]](#)). To investigate in greater detail whether cells competent for transcellular infection are selectively kept away from transiting to mitosis, we first used a triple fluorescent sensor enabling to track all cell-cycle phases *in planta*. Originally developed for *Arabidopsis*, the plant cell-cycle indicator (PlaCCI) combines a component of the pre-replication complex (CDT1a), the replicative histone variant (H3.1) and the N-terminal domain of a B-type cyclin (N-CYCB1;1) as G1-, S- and late G2-phase reporters, respectively (**Figure 4A** [\[9\]](#); Desvoves et al., 2020 [\[10\]](#); Echevarria et al., 2021 [\[11\]](#)). The N-terminal domain of CYCB1;1 contains a destruction box (D-box) ensuring a rapid turnover of the protein after one mitotic cell cycle. These cell-cycle markers are expressed at early stages of organ development but disappear in terminally differentiated cells (Desvoves et al., 2020 [\[10\]](#)). When expressing the PlaCCI construct in transgenic *Medicago* roots, all three markers were successfully visualized in reactivated cortical cells upon nodule primordium formation (**Figure 4B–C** [\[12\]](#)). Cortical cells passed by an infection thread kept low but detectable amounts of the *Arabidopsis* H3.1 variant, indicating that they transited through S-phase (**Figure 4C** [\[13\]](#)). However, these cells stopped before accumulating substantial levels of N-CYCB1;1, in contrast to their direct neighbours. We also confirmed latter observations when using an *Arabidopsis* CYCB1;2-based fluorescent reporter (**Figure 4D** [\[14\]](#)), labelling only cells that transit through G2/M (**Figure 4** [\[15\]](#)–figure supplement 1A; Weimer et al., 2016 [\[16\]](#)). Together, these results verified a selective decrease in mitotic competence occurring on the cortical infection thread trajectory (Yang et al., 1994 [\[17\]](#)) as rhizobia pass the middle cortical layer (C3) in *Medicago* roots. Importantly, we did not observe the G1-phase marker accumulating in cells undergoing infection (**Figure 4C** [\[18\]](#)), favouring the hypothesis of a cell-cycle arrest and exit decision (**Figure 2E** [\[19\]](#)) rather than a rapid switch to another endoreduplication round (Desvoves et al., 2019 [\[20\]](#) – pre-print).

We next tested whether transcriptional reporters of a target gene necessary to execute cell division get selectively repressed on the path of a cortical infection. The late G2/M gene *KNOLLE* codes for a syntaxin protein specialized in angiosperm cytokinesis and is oppositely controlled when cells do (Haga et al., 2007 [\[21\]](#)) or shall not (Takahashi et al., 2019 [\[22\]](#)) commit to mitosis. The transcription of *KNOLLE* and other late cell-cycle genes (e.g., *CYCB1;2*) in *Arabidopsis* is primarily regulated by mitosis-specific activator (MSA) cis-elements present in their promoters that are targeted by activator- and repressor-type three Myb repeats (MYB3R) transcription factors (Haga et al., 2007 [\[23\]](#); Kobayashi et al., 2015 [\[24\]](#)). Mutations in all three MSA core sequences present in the *Medicago* *KNOLLE* promoter significantly reduced the expression of a β -glucuronidase (*GUS*) reporter gene in *Nicotiana benthamiana* leaf cells in the presence of a hyperactive form of a MYB3R transcriptional activator (NtmybA2 Δ 630; **Figure 4** [\[25\]](#)–figure supplement 1B; Araki et al., 2004 [\[26\]](#)), suggesting that the *Medicago* *KNOLLE* gene can be used as a *bona fide* late G2/M readout. Using the pKNOLLE::NLS-3xVenus construct, we found that the activation of the *KNOLLE* promoter

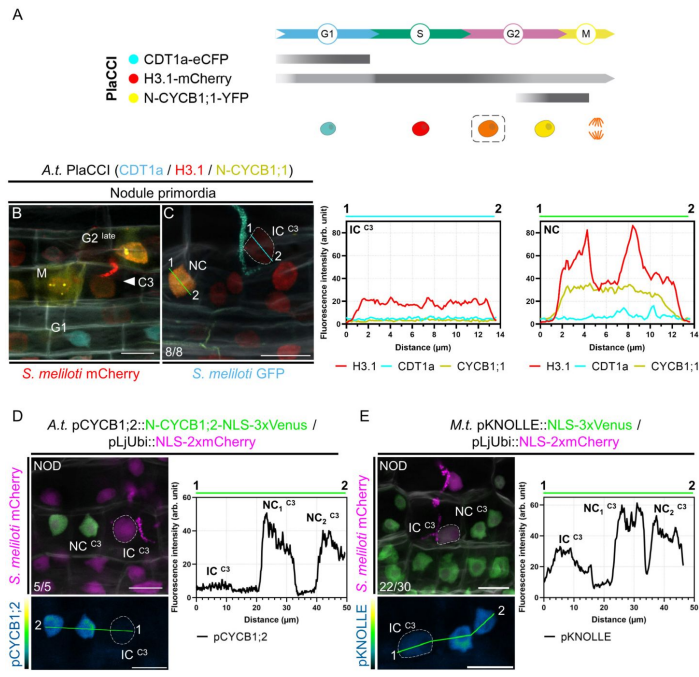


Figure 4.

A tight control over host cells' mitotic commitment enables passage of the future nodule meristem

(A) Schematic representation of Arabidopsis CDT1a-eCFP, H3.1-mCherry and N-CYCB1;1-YFP distribution (horizontal bars) and fluorescence intensity (grey saturation) throughout the different cell-cycle phases (adapted from Echevarria et al., 2021; see also Desvoyes et al., 2020). CDT1a-CFP accumulates in G1 and is rapidly degraded during the G1/S transition (blunt end bar). H3.1 is predominantly expressed during S-phase and incorporated during DNA replication. N-CYCB1;1-YFP is present in late G2 and mitotic cells and is completely degraded in anaphase (blunt end bar). Nuclei of increasing size and DNA content are coloured according to their CDT1a (cyan), H3.1 (red) and N-CYCB1;1 (yellow) content in/around decondensed chromatin (G1, S, G2) or condensed chromosomes (M). H3.1 levels decrease in differentiating cells (dotted-line box). (B-C) Confocal images of whole-mount Medicago WT roots expressing the Arabidopsis PlaCCI reporter in nodule primordia at 7 dpi with mCherry-(B) or GFP-producing *S. meliloti* (C). Images are maximum intensity projections and show merged fluorescent channels (eCFP and GFP: cyan; mCherry: red; YFP: yellow; Calcofluor white cell-wall staining: grayscale). (B) The filled arrowhead points to a cell of the C3 layer just penetrated by an infection thread. Cell-cycle phases of non-infected neighbouring cells are indicated. (C) Left panel: the dotted-line ROI indicates the nucleus of an infected cell (IC) from the C3 layer being passed by a cortical infection thread. Numbers indicate the frequencies of observation of the absence of CDT1a- or N-CYCB1;1-associated signals in the nucleus of the last infected cortical cell. Right panels: fluorescence intensity profiles of CDT1a-, H3.1- and N-CYCB1.1-associated signals along the cyan and green transects shown in (C). Scale bars: 20 μ m. (D) Left panels: confocal images of a whole-mount WT root co-expressing a destabilized triple-Venus nuclear reporter driven by the Arabidopsis *CYCB1;2* promoter (pCYCB1;2::N-CYCB1;2-NLS-3xVenus) and a nuclear-localized tandem-mCherry as a transformation marker, 12 dpi with mCherry-producing *S. meliloti*. Numbers indicate the frequencies of observation of the absence of the triple-Venus reporter signal in the nucleus of the last infected cortical cell. Two independent transformation experiments were performed with 3 to 4 composite plants analyzed per replica. (E) Left panels: confocal images of a whole-mount WT root co-expressing a transcriptional reporter of Medicago *KNOLLE* driving a nuclear-localized triple-Venus (pKNOLLE::NLS-3xVenus) together with a nuclear-localized tandem-mCherry as a transformation marker, 12 dpi with mCherry-producing *S. meliloti*. Numbers indicate the frequencies of observation of nodule primordia where the triple-Venus reporter signal is kept comparably low on the cortical infection thread trajectory. A total of 12 composite plants from 2 independent transformation experiments were analyzed. (D-E) Images are maximum intensity projections and show merged fluorescent channels (Venus: green; mCherry: magenta; Calcofluor white cell-wall staining: grayscale in the upper panels). The Venus channel is shown in Green Fire Blue when isolated in lower panels, with blue or yellow indicating low or high fluorescence levels, respectively. The dotted-line ROIs indicate nuclei of infected cells (IC) from the C3 layer being passed by a cortical infection thread. The fluorescence intensity profiles of pCYCB1;2 and pKNOLLE reporter-associated signals along the green transects are shown in the corresponding right panels. NC: neighbouring cell. NOD: nodule primordium. Scale bars: 20 μ m.

was lower on the cortical infection thread trajectory in comparison to neighbouring, proliferating cells from nodule primordia (**Figure 4E**). An equivalent construct, ensuring the timely elimination of fluorescent reporters after one canonical cell cycle (pKNOLLE::D-box-3xVenus-NLS; **Figure 4**–figure supplement 1C), corroborated a selective control of the G2/M transition in cells undergoing transcellular infection. Combining our observations of enlarged nuclei (**Figure 2C**), the absence of a G1 marker and the reduced H3.1 levels and expression of mitotic genes (**Figure 2D** and **Figure 4B–E**), we conclude that the initial cortical cells along the infection thread trajectory conduct a single cell-cycle round and stay in a post-replicative (*i.e.*, G2) phase as long as they are crossed.

We further took advantage of the fact that the PlaCCI sensor provides a full view on cell-cycle phase progression in root tissues, in *Arabidopsis* (Desvoyes et al., 2020) and *Medicago* (this study). CDT1a- and CYCB1;1-associated signals in the root tip enabled us to identify 2C (*i.e.*, G1) and 4C (*i.e.*, G2) nuclei, respectively (**Figure 4**–figure supplement 2A) – where 2C and 4C chromatin-values correspond to the DNA content of a basic diploid genome. Measuring the area of nuclei exhibiting the CYCB1;1-YFP signal in cortical layers of the root apical meristem (root tip) or the nodule primordium revealed that they were on average 1.57 or 1.63 times larger than the ones producing CDT1a-CFP, respectively (**Figure 4**–figure supplement 2B). This is in agreement with an increased DNA content as cells transit from the G1- to the late G2-phase. Surprisingly, proliferating nodule cells contained significantly wider nuclei than their diploid counterparts in the proliferation zone of the root apical meristem (**Figure 4**–figure supplement 2B), suggesting a higher ploidy level early on. These results lead to the intriguing possibility that reactivated inner cortical cells in *Medicago* enter the mitotic cell cycle with an increased amount of genetic material.

Proliferation of endopolyploid cortical cells is a distinctive feature of *Medicago* root nodules

Our data suggest that the dividing cortical cells contributing to nodule primordium formation are polyploid – *i.e.*, that they contain more than the two genome copies of a diploid (2n) cell, where “n” refers to the number of separate chromosomes. To assess this further, we identified and used the centromere-specific histone H3 variant (CENH3) from *Medicago* as a marker of individual chromosomes. Such fluorescently-tagged CENH3 proteins have already been successfully applied for chromosome quantification *in planta* (Lermontova et al., 2006; De Storme et al., 2016). Aligning the *Medicago* CENH3 amino acid sequence (G8A083) with characterized CENH3 sequences from *L. japonicus*, *Arabidopsis*, *H. vulgare* and *Oryza sativa* revealed the presence of canonical CENH3 features (**Figure 5**–figure supplement 1), including a variable N-terminal tail and the centromere targeting domain (CATD). When expressing a mCitrine-CENH3 fusion protein under the control of its own promoter (pCENH3::mCitrine-CENH3), CENH3 labelled up to 16 nuclear foci and localized exclusively at the presumed centromeres during mitosis (**Figure 5A and B**) in diploid root tip cells. As the signal was rather weak, we scored centromeric foci in roots ectopically expressing the protein (pUbi::mCitrine-CENH3; **Figure 5A**). Here, a reliable signal was observed in an increased proportion of nuclei (82%) displaying not more than 14 to 16 mCitrine-CENH3 foci (versus only 55% when CENH3 was expressed under its endogenous promoter; n = 55; **Figure 5B**) – reflecting the base ploidy in *M. truncatula* with 2n = 16. We therefore used the pUbi::mCitrine-CENH3 reporter construct for our further analyses.

In a first set of experiments, we used CENH3 to discriminate between polyploid cells undergoing endomitosis or endocycling. In endomitosis, entire chromosome replication and partial mitotic progression result in a doubling of the chromosome / centromeric foci number (Iwata et al., 2011; De Storme et al., 2016). By contrast, endocycling cells undergo repeated rounds of DNA synthesis without segregating the newly replicated parts (Edgar et al., 2014). Here, copied chromosome arms do not condense and spatial separation of sister kinetochores does not occur, so that the number of compact CENH3 signals corresponds to the initial number of chromosomes (up to 8C; **Figure 5**–figure supplement 2A; Lermontova et al., 2006). As the occurrence of

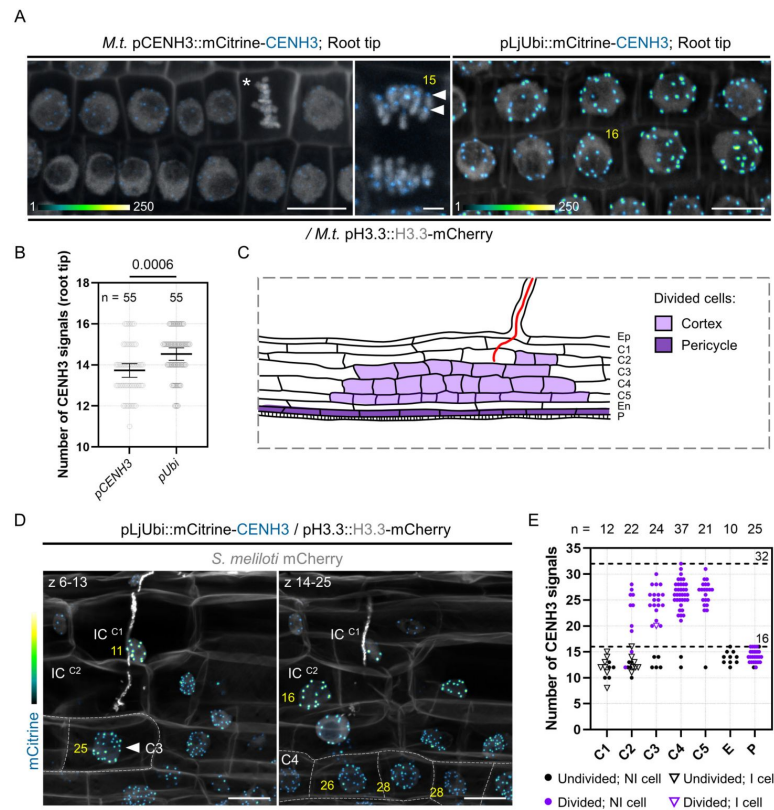


Figure 5.

Dividing cortical cells initiating Medicago nodule primordium formation are tetraploid

(A) Confocal images of whole-mount WT transgenic roots expressing mCitrine-CENH3 under the control of native (*M.t.* pCENH3; left and middle panels) or constitutive promoters (pLjUbi; right panel). Simultaneous expression of H3.3-mCherry enables the recognition of condensed (star) and segregating chromosomes (middle panel). mCitrine-CENH3 labels the centromeric region of individual chromosomes (filled arrowheads). The number of CENH3-labeled foci determined across image stacks is indicated in yellow. Images are maximum intensity projections and show merged fluorescent channels (mCitrine: Green Fire Blue; mCherry and Calcofluor white cell-wall staining: grayscale). In the Green Fire Blue colour scheme, blue or yellow indicate low or high fluorescence levels, respectively (min. to max. = 1-140 in the middle panel). Scale bars: left and right panels = 10 μ m; middle panel = 2 μ m. (B) Quantification of the number of centromeric signals in transgenic root tips expressing mCitrine-CENH3 under native (pCENH3) or constitutive (pUbi) promoters ($n = 55$ nuclei). All data points are shown and are from 2 (pUbi) to 3 (pCENH3) independent experiments with 7 composite plants analyzed per construct. Horizontal bars indicate sample means with 95% confidence interval. Differences were statistically significant (p -value = 0.0006) using a Mann-Whitney test. (C) Schematic representation of an early nodule primordium where anticlinal cell divisions occurred in the pericycle (P; dark violet) and cortical cells (C2 to C5; light violet). Rhizobia inside the infection thread are depicted in red. Ep: epidermis. En: endodermis. (D) Maximum intensity projections of an early nodule primordium at the developmental stage represented in (C), formed in a WT root expressing the pLjUbi::mCitrine-CENH3 / pH3.3::H3.3-mCherry construct at 7 dpi with mCherry-producing *S. meliloti*. Dotted-line ROIs indicate the contours of divided cells in the inner (C4) and middle (C3) cortical layers. The number of CENH3-labeled foci, determined across image stacks and indicated in yellow are: 11 and 16 in undivided, infected cells (IC) from the outer cortical layers (C1; left panel and C2, right panel) and 25 to 28 in divided, uninfected cells from the C3 (filled arrowhead; left panel) and C4 layers (right panel). Images show merged fluorescent channels (mCitrine: Green Fire Blue; mCherry and Calcofluor white cell-wall staining: grayscale). In the Green Fire Blue colour scheme, blue or yellow indicate low or high fluorescence levels, respectively. Scale bars: 20 μ m. (E) Quantification of the number of centromeric signals in early nodule primordia at the developmental stage depicted in (C), formed in inoculated WT roots (7 to 12 dpi) constitutively expressing mCitrine-CENH3. The number (n) of analyzed nuclei in each cell layer is indicated on the top. E: endodermis. P: pericycle. All data points are shown with black or violet symbols indicating undivided or divided cells, respectively. NI: non-infected cell (discs). I: infected cell (triangles). Horizontal dotted lines are positioned at $y = 16$ and $y = 32$, corresponding to diploid ($2n = 16$) or tetraploid ($4n = 32$) cellular states in Medicago. Data are from 2 independent experiments with 7 nodule primordia from 5 composite plants analyzed.

endocycling cells has been previously reported in tissues infected by AM fungi, we first tested the CENH3-reporter in roots inoculated with *Rhizophagus irregularis*. Indeed, cortical cells entered by the fungus displayed large nuclei with no more than 16 labelled foci (**Figure 5** [figure supplement 2B](#)), supporting their endocycling nature ([Carotenuto et al., 2019a](#) [figure supplement 2B](#); [Lermontova et al., 2006](#) [figure supplement 2C](#)). Up to 16 CENH3 foci were also observed in diploid cortical cells from the root tip and pericycle cells from lateral root primordia (**Figure 5** [figure supplement 2C](#)). However, the majority of nodule cells exhibited between 26 and 32 fluorescent centromeric foci (75,5% in nodule primordia from stage IV to VI, $n = 106$; **Figure 5** [figure supplement 2B–C](#)) and a visibly increased number of chromosomes on mitotic figures (**Figure 5** [figure supplement 2B](#)). These observations strongly support previously published data proposing that proliferating cortical cells giving rise to Medicago root nodules are tetraploid ($4n = 32$; [Torrey & Barrios, 1969](#) [figure supplement 2B](#)).

We next tested whether endopolyploid cells are already present at an earlier stage of nodule development, where cell division events are confined to the pericycle, the inner cortical cells (C4, C5) and occasionally occur in the third (C3) and second (C2) cortical layers (**Figure 5C** [figure supplement 2B](#); [Xiao et al., 2014](#) [figure supplement 2B](#)). Indeed, cells originating only from cortical cell divisions (C2 to C5) showed an increased number of chromosomes (**Figure 5D–E** [figure supplement 2B](#)) compared to recently divided cells originating from the pericycle (**Figure 5E** [figure supplement 2B](#)). Notably, cells performing anticlinal divisions in the middle cortex (**Figure 5D** [figure supplement 2B](#), left panel) that will later be crossed by the infection thread (**Figure 5E** [figure supplement 2B](#), violet triangle) were also polyploid. From these observations, we concluded that a key feature of cortical cells in Medicago nodules is their ability to segregate a higher number of chromosome copies - corresponding to 8C in a tetraploid cell entering mitosis.

Infected cells within the nodule primordium selectively halt their preparations for mitosis

In a next set of experiments, we studied to which extent the cells positioned on the infection thread trajectory (C3 to C4/5) prepare for mitosis. For this, we made use of the pre-mitotic loading characteristics of CENH3 to the centromeres, which mainly occurs during the G2 phase and prior to the spatial separation of sister kinetochores (**Figure 6A** [figure supplement 1A–B](#); [Lermontova et al., 2006](#) [figure supplement 1A–B](#), [2007](#) [figure supplement 1A–B](#), [2011](#) [figure supplement 1A–B](#)). In proliferating populations of the root tip (**Figure 6** [figure supplement 1A–B](#)) and nodule primordia (**Figure 6A–B** [figure supplement 1A–B](#)), late G2 cells contained large nuclei where the majority of sister kinetochores was split and appeared as twin foci. We further noticed that centromere separation in Medicago is a gradual rather than a simultaneous process, as root tip cells with comparable nuclear areas often exhibited an intermediate number of fluorescent doublets (**Figure 6** [figure supplement 1A](#)) of similar appearance and length on a single focal plane (**Figure 6** [figure supplement 1B–D](#)). To address the question to which extent cortical cells that are competent for infection progress through the G2 phase and commit to nuclear division, we assessed the gradual formation of sister centromeres. Although the nuclear size of infected C3 cells suggested an 8C DNA content, as found in late G2-phase cells of the nodule tissue (**Figure 6B, C** [figure supplement 1A–B](#)), none of these infected cells showed the typical CENH3 twin foci formed when sister kinetochores split ($n = 13$; **Figure 6B** [figure supplement 1A–B](#) and **D** [figure supplement 1A–B](#)). Such control over mitotic commitment was tightly maintained in the majority of infected, C4/5-derived cells (75,5%; $n = 53$; **Figure 6D** [figure supplement 1A–B](#)) which did not display any visible sign of centromere separation across image stacks (**Figure 6B** [figure supplement 1A–B](#)), despite a presumably duplicated genomic DNA content (**Figure 6C** [figure supplement 1A–B](#)). A smaller proportion of the recently infected cells (20,7%) partially prepared for chromosome segregation and showed up to 3 fluorescent doublets (**Figure 6D** [figure supplement 1A–B](#)) of comparable lengths to those measured in G2-phase progressing cells (**Figure 6E** [figure supplement 1A–B](#)), whereas only two cells (3,8%) with internalized bacteria got closer to nuclear and likely to cell division (**Figure 6B** [figure supplement 1A–B](#) and **D** [figure supplement 1A–B](#)). Together, by combining several marker strategies based on histone H3 dynamics and on *bona fide* mitotic reporters, we conclude that most middle (C3) and inner (C4/5-derived) cortical cells competent for intracellular infection enter a post-replicative, G2-type phase without proceeding towards M-phase. Endopolyploid cortical cells hosting infection threads have presumably gained the ability to execute a differentiation program rather than entering cell division at the end of G2.

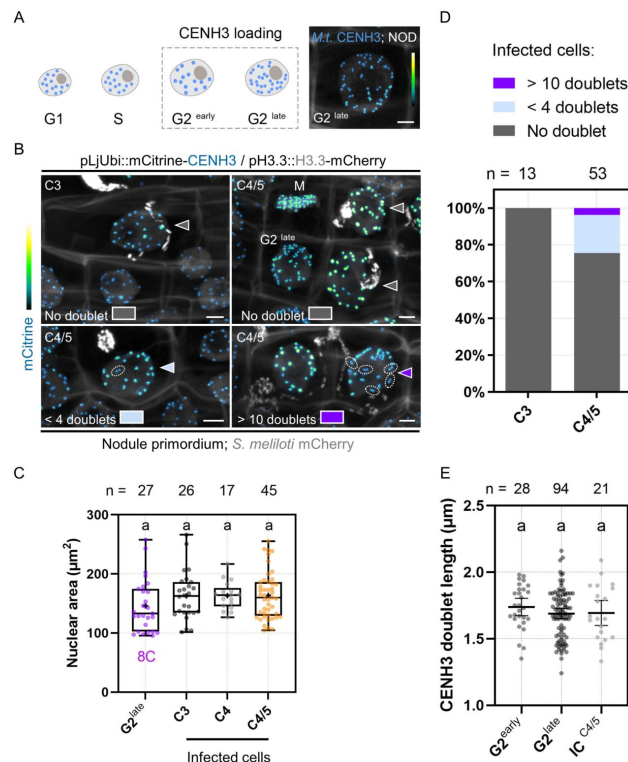


Figure 6.

Primary infected nodule primordium cells do not reach full competence for chromosome segregation

(A) Schematic representation of CENH3 deposition at centromeres (blue dots) throughout the different pre-mitotic (G1, S, G2 early and G2 late) cell-cycle phases in plants. Nuclei of increasing size and DNA content are depicted as ovals. The dotted-line frame indicates the timing of CENH3 loading at replicated centromeres, occurring during G2 before sister kinetochores (blue doublets) split in preparation to mitosis (Lermontova et al., 2007). Right panel: confocal image of a late G2-phase cell observed in a nodule (NOD) in an inoculated WT transgenic root (12 dpi) constitutively expressing mCitrine-CENH3. The majority of fluorescent signals appear as doublets corresponding to sister kinetochores. Scale bar: 5 µm. (B) Confocal images of whole-mount WT roots expressing the pLjUbi::mCitrine-CENH3 / pH3.3::H3.3-mCherry construct and inoculated with mCherry-producing *S. meliloti* (7 to 12 dpi). Filled arrowheads point to nuclei of infected nodule primordium cells. Dotted-line ROIs indicate mCitrine-CENH3 doublets appearing as twin spots on the same focal plane. Neighbouring cells in late G2 or undergoing mitosis (M) are indicated. The presence of CENH3-labeled twin spots was assessed across image stacks in the nuclei of infected cells from the middle (C3) and inner (C4/5) cortical layers and colour-coded as follows: grey = no doublet (upper panels); light blue = 1 to 3 doublets (lower left panel); dark violet = 11 or more doublets (lower right panel). Scale bars: 5 µm. (A-B) Images are maximum intensity projections (mCitrine: Green Fire Blue; mCherry and Calcofluor white cell-wall staining: grayscale). In the Green Fire Blue colour scheme, blue or yellow indicate low or high fluorescence levels, respectively. (C) Quantification of the nuclear area at the equatorial plane in cells being passed by a cortical infection thread (infected cells) in nodule primordia formed by WT transgenic roots inoculated with *S. meliloti* (7 to 12 dpi). The number (n) of analyzed nuclei in each cell layer (C3 to C4/5) is indicated on the top. The 8C chromatin-value is given for uninfected nodule primordium cells in late G2 showing close to 32 mCitrine-CENH3 doublets as identified in (A) and (B). All data points are shown and crosses indicate sample means. Differences were not statistically significant according to a Kruskal-Wallis test followed by Dunn's multiple comparisons test. At least 9 composite plants from 2 independent experiments were analyzed. (D) Quantification of infected cells in C3 (n = 13) and C4/5 (n = 53) cortical layers in nodule primordia, showing no doublet (grey), 1 to 3 doublets (light blue) or more than 11 doublets (dark violet) labelled by mCitrine-CENH3. Nine composite plants from 2 independent experiments were analyzed. (E) Quantification of the length of mCitrine-CENH3 doublets appearing as twin spots on the same focal plane in early G2 (n = 12), late G2-phase cells (IC) from the C4/5 cortical layer (n = 13) in nodule primordia. n = 28 (G2^{early}), 94 (G2^{late}) and 21 (IC C4/5) doublets. Horizontal bars indicate sample means with 95% confidence interval. Differences were not statistically significant according to a Kruskal-Wallis test followed by Dunn's multiple comparisons test. All data points are shown and are from 5 to 7 composite plants from 2 independent transformation experiments.

The symbiosis-specific NF-YA1 subunit holds potential to modulate the G2/M transition

Our data suggest that cell-cycle progression and bacterial infection are highly coordinated, although the precise molecular players able to redirect cell-cycle decisions while promoting cortical infections remain to be identified. Prominent candidates are NF-YA1 proteins that were evolutionary recruited to RNS, where they sustain both organogenesis and intracellular infection in a range of nodulating legume and non-legume species (Combiér et al., 2006 [DOI](#); Laporte et al., 2014 [DOI](#); Xiao et al., 2014 [DOI](#); Hossain et al., 2016 [DOI](#); Bu et al., 2020 [DOI](#)). NF-YA1 plays an unambiguous role as a positive regulator of cell divisions (Soyano et al., 2013 [DOI](#)) but was also recently shown to function in the early differentiation and specification of nodule cells in *Medicago* (Schiessl et al., 2023 [DOI](#) – pre-print). This prompted us to assess whether NF-YA1 could act as a switch at mitotic entry. To test this, we used WT NF-YA1 and a mutated variant lacking a functional DNA-binding domain (mutDBD) as eGFP fusion proteins. As expected, both variants localized to the nucleus in *N. benthamiana* epidermal cells (Figure 7A [DOI](#); Laporte et al., 2014 [DOI](#)). To foster cell cycle re-entry in such differentiated cells, we ectopically expressed the Arabidopsis D-type cyclin *CYCD3;1* in mature leaves as described in the recently developed ‘cell division-enabled leaf system’ (CDELS; Xu et al., 2020 [DOI](#)). As these re-activated cells commit to mitosis and perform cytokinesis (Figure 7B [DOI](#)), they also enable the activation of promoters containing MSA cis-elements (Xu et al., 2020 [DOI](#)) as found in the *Medicago KNOLLE* promoter (Figure 4 [DOI](#)–figure supplement 1B). We confirmed this by expressing eGFP-*KNOLLE* from its endogenous promoter in the absence and presence of Arabidopsis *CYCD3;1*. The *KNOLLE* fusion protein was only detected when co-expressed with *CYCD3;1* using immunoblot analyses (Figure 7 [DOI](#)–figure supplement 1A) and labelled newly formed cell-division planes (Figure 7 [DOI](#)–figure supplement 1B). This is consistent with previous reports showing that the syntaxin *KNOLLE* accumulates only in dividing cells. We also confirmed the activation of the *Medicago KNOLLE* promoter (p*KNOLLE*::GUS) in CDEL samples ectopically expressing *CYCD3;1* by using a fluorometric GUS assay (Figure 7C [DOI](#)). Next, we tested the impact of NF-YA1 on the activity of promoters driving cell cycle-dependent gene transcription in the CDEL samples. While the presence of NF-YA1 did not alter *GUS* expression when driven from the histone *H4* promoter (p*H4*::GUS) that is active during DNA synthesis (Figure 7C-D [DOI](#); Yang et al., 1994 [DOI](#)), it significantly repressed p*KNOLLE*::GUS activation (Figure 7E [DOI](#)). This effect was fully abolished when using the mutated NF-YA1^{mutDBD} variant (Figure 7E [DOI](#)). Interestingly, the repressive effect of NF-YA1 was stronger compared to the one observed when using a MYB3R transcriptional repressor from *Medicago* (Figure 7 [DOI](#)–figure supplement 2A–B) known to halt the expression of late G2/M genes (Haga et al., 2007 [DOI](#); Takahashi et al., 2019 [DOI](#)). These results suggest that the NF-YA1 DNA-binding subunit prevents an unscheduled entry into mitosis in de-differentiating leaf cells, despite the action of ectopically-expressed *CYCD3;1*.

We then tested whether such effect could be observed at the cellular level in the CDEL system. For this, we ectopically co-expressed non-fluorescently-tagged *CYCD3;1* and NF-YA1 and additionally transformed these leaves with fluorescently tagged *Medicago* H3.3 and *KNOLLE* proteins, expressed from their native regulators (Figure 7 [DOI](#)–figure supplement 1B). While the constitutive presence of NF-YA1 did not prevent cell division events *per se*, a significantly lower proportion of transformed cells produced detectable levels of *KNOLLE* (Figure 7 [DOI](#)–figure supplement 1C), indicating that less pavement cells engaged in mitosis at the investigated experimental timepoint. Cells visibly accumulating the *KNOLLE* protein still progressed through M-phase in the presence of NF-YA1 (Figure 7 [DOI](#)–figure supplement 1B), although a smaller proportion of them fully completed cell division at 64 hours when compared to the control samples (Figure 7 [DOI](#)–figure supplement 1D). From these data we concluded that NF-YA1 is able to control mitotic entry in de-differentiating, endoreduplicated cells at least in the CDEL system. Since symbiotically infected middle and inner primordium layers, where NF-YA1 expression was found to be confined in

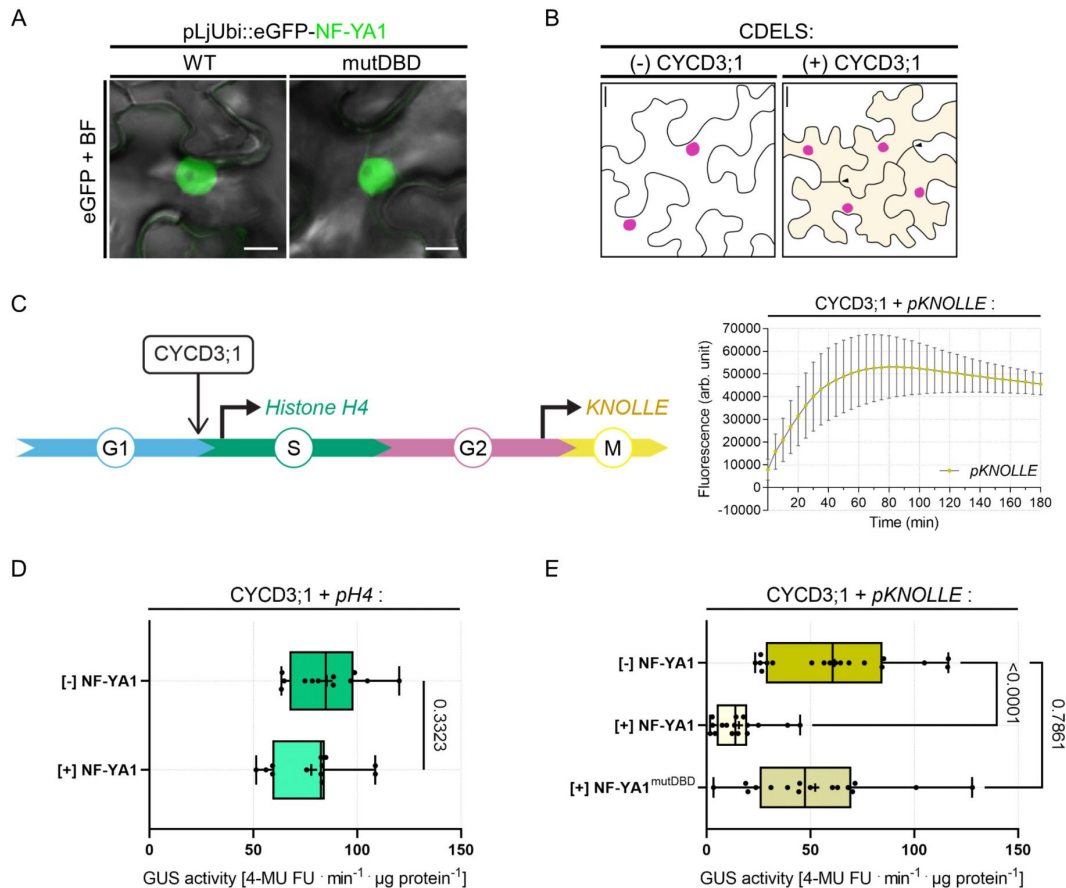


Figure 7.

NF-YA1 specifically reduces the expression of a G2/M gene in a cell-division enabled system

(A) Live-cell confocal images of *N. benthamiana* epidermal cells ectopically expressing eGFP-NF-YA1 variants: WT (left panel) or impaired in DNA recognition (mutDBD; right panel). Both fusion proteins accumulate in the nucleus and were used in fluorometric β -glucuronidase (GUS) assays. Images are maximum intensity projections and show merged fluorescent (eGFP: green) and bright field (BF) channels. Scale bars: 10 μ m. (B) Schematic representation of the cell division-enabled leaf system (CDELS) on the scale of epidermal pavement cells in *N. benthamiana*. Differentiated cells in mature leaves (left panel) re-enter the canonical cell cycle upon ectopic expression of the Arabidopsis D-type cyclin CYCD3;1 (right panel). Nearly all re-activated pavement cells (diffuse yellow colouring) have completed cytokinesis after 3 days (filled arrowheads). See also [Figure 7](#) – figure supplement 1B. Scale bars: 20 μ m. (C) Left panel: schematic representation of the CDEL system on the scale of the canonical cell cycle. Ectopically-expressed CYCD3;1 targets and activates cyclin-dependent kinases (CDK, not shown), fostering the G1/S transition. Progression throughout the different cell-cycle phases is accompanied by the sequential activation of DNA synthesis (S) and mitotic genes (M). Histone *H4* and *KNOLLE* promoter-reporters were selected as readouts for G1/S and late G2/M transcriptional waves, respectively. Right panel: fluorometric GUS assay in tobacco leaf cells ectopically expressing CYCD3;1 and the *Medicago* pKNOLLE::GUS (*pKNOLLE*) reporter construct. Fluorescence curves (GUS-mediated hydrolysis of 4-MUG) over time are shown for 3 biological replicates. Error bars indicate standard deviation. Data are from 1 of 4 independent transformation experiments. (D) Activity of GUS driven by the histone *H4* promoter (*pH4*) in the absence or presence of ectopically-expressed eGFP-NF-YA1 (NF-YA1) in CDEL samples (+ CYCD3;1). All data points are shown and crosses indicate sample means. Differences were not statistically significant (p -value = 0.3323) according to an unpaired *t* test with Welch's correction. Data are from 3 independent transformation experiments. n = 12 ([-] NF-YA1) and 12 ([+] NF-YA1) biological samples. (E) Activity of GUS driven by the *KNOLLE* promoter (*pKNOLLE*) in CDEL samples (+ CYCD3;1) in the absence or presence of ectopically-expressed eGFP-NF-YA1 variants with WT (NF-YA1) or mutated DNA-binding domain (NF-YA1^{mutDBD}). All data points are shown and crosses indicate sample means. Statistically significant differences ([-] NF-YA1 versus [+] NF-YA1; p -value < 0.0001) or non-significant differences ([-] NF-YA1 versus [+] NF-YA1^{mutDBD}; p -value = 0.7861) are based on a Kruskal-Wallis test followed by Dunn's multiple comparisons test. Data are from 4 to 5 independent transformation experiments. n = 18 ([-] NF-YA1), 16 ([+] NF-YA1), 16 ([+] NF-YA1^{mutDBD}) biological samples.

(Laporte et al., 2014 [↗](#); Liu et al., 2019b [↗](#)), are comprised of endopolyploid cells (**Figure 5C–E** [↗](#)), we hypothesize that the Medicago NF-YA1 protein might similarly modulate cell-cycle progression during symbiotic infections of legume roots by rhizobia.

Discussion

Failure to timely coordinate infection and organogenesis programs in the root cortex results in the formation of poorly colonized nodules that are unable to support the plant's demand to obtain fixed nitrogen (Guan et al., 2013 [↗](#); Laporte et al., 2014 [↗](#); Schiessl et al., 2023 [↗](#)–pre-print). Here, we sought to understand at cellular resolution, whether cell-cycle progression and cellular reprogramming are altered in a spatially confined array of host cells that gain a time-limited stage of competence for transcellular infection.

Our data support a model where dividing cells contributing to nodule formation, including target cells for cortical infection, are polyploid and contain 4 copies (C) of each chromosome (**Figure 5D–E** [↗](#)). In Medicago, these 4C cells either pre-exist in the root cortex (Vinardell et al., 2003 [↗](#); Carotenuto et al., 2019a [↗](#); Carotenuto et al., 2019b [↗](#)) or arise from a single endoreduplication round prior to their possible entry into cell division (Libbenga & Torrey, 1973 [↗](#)). As competent cells are crossed by a cortical infection thread or fully internalize bacteria and despite being embedded into a proliferating tissue, they reach an 8C DNA content (**Figure 6C** [↗](#) and **Figure 8** [↗](#)) but without committing to a subsequent cell division (**Figure 4C–E** [↗](#)). Rather, cells supporting transcellular infections presumably use a cell-cycle GAP phase after DNA replication (*i.e.*, G2) and extend such phase to remodel their histone H3 composition (**Figure 2A–B** [↗](#) and **D** [↗](#)). Finally, this differentiation process and concomitant transcellular passage are accomplished within a single, last cell-cycle round (**Figure 2E** [↗](#); Otero et al., 2016 [↗](#)). Here, and similar to infected root hairs (Breakspear et al., 2014 [↗](#)), a rapid switch to another endocycle is likely prevented as both G1- and S-phase markers are kept low (**Figure 4C** [↗](#)). Taken together, these events control a cell-cycle progression state where molecular players operate to ensure a successful transmission of rhizobia to the first nodule cells.

Modulating endopolyploidy during rhizobial infections

In line with findings from Arabidopsis reporting that a downregulation of *H3.1* gene expression and a massive eviction process contribute to various differentiation programs, *e.g.*, in the male and female gametes (Ingouff et al., 2010 [↗](#)), the root meristem, the embryo (Otero et al., 2016 [↗](#)) and the female germline precursor (Hernandez-Lagana & Autran, 2020 [↗](#)), we show that the genetic program enabling intracellular infections of Medicago roots accesses the same mechanism (**Figure 2A–B** [↗](#) and **Figure 2** [↗](#)–figure supplement 1C–E). Reciprocally, our data indicate that high H3.1 levels tend to be maintained in infected cells with blocked or aborted infections (**Figure 3A and C** [↗](#) and **Figure 3** [↗](#)–figure supplement 1A–D). Furthermore, the nuclear enlargement in infected cells (**Figure 2C** [↗](#)) indicates profound transcriptional reprogramming and activation of new gene regulatory networks, as previously shown to occur in LCO-treated (Knaack et al., 2022 [↗](#)) or young symbiotic cells (*i.e.*, infected root hairs and 8C cells from mature nodules; Breakspear et al., 2014 [↗](#); Nagymihály et al., 2017 [↗](#); Liu et al., 2019a [↗](#)). We expanded the latter view by showing that tetraploid cortical cells passed by an infection thread also reach a presumable 8C DNA content (**Figure 6C** [↗](#)). As reported for growing cell types in Arabidopsis (Bhosale et al., 2018 [↗](#)), such alteration in endopolyploidy might support, among others, changes in cell wall composition, which have also recently been demonstrated to occur specifically around infected cells in Medicago (Su et al., 2023 [↗](#); Gaudioso-Pedraza et al., 2018 [↗](#)). In the determinate nodulator *L. japonicus*, polyploidization of early nodule cells might be controlled by *VAGRANT INFECTION*

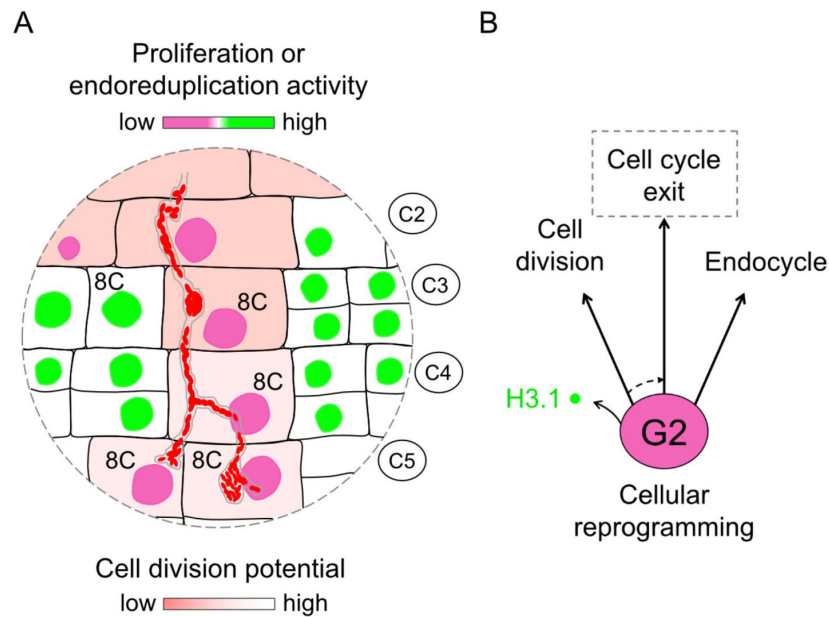


Figure 8.

Model illustrating characteristic cellular traits on the cortical infection thread trajectory

(A) Cortical cells competent for sustained transcellular infection thread progression are characterized by a reduced proliferation or endoreduplication activity (pink to green two-colour gradient). By the time they are crossed and ultimately internalize bacteria, such tetraploid cells (this study, [Torrey and Barrios, 1969](#)) reach a post-replicative, 8C ploidy stage but contrary to direct neighbour cells, they do not commit to a subsequent cell division (red to white two-colour gradient). (B) Rather, cells supporting transcellular infection presumably enter a prolonged G2-phase (this study, [Yang et al., 1994](#)) during which they remodel their histone H3 composition as part of their cellular reprogramming for infection. This differentiation process and concurrent transcellular passage are accomplished within a last cell-cycle round. We propose that cellular factors controlling the G2/M transition and eventually the exit from the canonical cell cycle (dashed arrow) are prominent candidates to support intracellular infection competence.

THREAD 1 (VAG1), a putative component of the DNA topoisomerase VI (topo VI) complex, as cortical cells with reduced ploidy levels cannot be infected in the *vag1* mutant (Suzaki et al., 2014 [↗](#)).

Interestingly, but different to infections by mycorrhizal fungi where repetitive rounds of endoreduplication occur (Russo & Genre, 2021 [↗](#)) and support fungal colonization (Figure 1C [↗](#); Carotenuto et al., 2019a [↗](#); Carotenuto et al., 2019b [↗](#)), markedly reduced histone H3.1 contents in cells maintaining rhizobial infections in the root cortex up to the first recipient cells in nodule primordia indicate reduced endocycling rates (Figure 2D-E [↗](#); Otero et al., 2016 [↗](#)). Such attenuation of cell-cycle progression together with decreased transcript levels of mitotic cyclins were shown to occur in budding yeast and Arabidopsis (Negishi et al., 2016 [↗](#); Gigli-Bisceglia et al., 2018 [↗](#)) following the activation of a cell-wall integrity (CWI) checkpoint. More recently, Ma and colleagues proposed a scenario where the CWI kinase THESEUS1 (THE1), with a related gene being induced in infected root hairs (Breakspear et al., 2014 [↗](#); Liu et al., 2019a [↗](#)), triggers cell wall stiffening as a response to endoreduplication attenuation (Ma et al., 2022 [↗](#)).

A G2-phase arrest and attenuated endoreduplication as a plausible context for cortical infections

Reaching and staying in a pre-mitotic stage has been proposed to facilitate the long-distance progression of infection threads in legume species forming so-called cortical, pre-infection threads (PITs) (Yang et al., 1994 [↗](#); Timmers et al., 1999 [↗](#); Foucher & Kondorosi, 2000 [↗](#)). These structures resemble the onset of a cell division plane that requires the formation of a cortical division zone, which is established from late S-through G2 phases (Costa, 2017 [↗](#)). Interestingly, this cell-cycle progression window coincides with the one we observed in infected cortical cells (Figure 2A-E [↗](#) and Figure 4A-C [↗](#)). In addition, the existence of pre-mitotic and G2-arrested cortical cells on the infection thread trajectory is in line with a recent report showing that a rhizobium-induced mitotic module, comprising an α -Aurora kinase gene and its MYB3R transcriptional activator, supports endosymbiotic infection in Medicago (Gao et al., 2022 [↗](#)). Interestingly though, overexpression of the activator-type MYB3R1 gene resulted in the formation of abnormally bulbous or branched infection threads (Gao et al., 2022 [↗](#)). These data emphasize the need to balance the activity of transcriptional regulators driving mitosis and cytokinesis in infected cells.

In summary, we hypothesize that endocycle suppression after a round of DNA replication (Breakspear et al., 2014 [↗](#)) and exiting the cell-cycle from the G2-phase might therefore create a permissive, yet controlled state for cortical infection thread progression, where polarization and cell-wall remodelling factors specialized for intracellular infection (Lace et al., 2023 [↗](#); Su et al., 2023 [↗](#)) act in a defined cell-cycle window to sustain infection thread growth and guidance across cortical layers.

Endopolyploidization following cytokinin perception: a legume innovation?

Using the centromeric histone H3 (CENH3) fluorescent reporter enabled us to assess the number of chromosomes, the progression through the G2 phase (Figure 6 [↗](#)–figure supplement 1A–B; Lermontova et al., 2006 [↗](#)) and to clearly identify diploid and polyploid cells (Figure 5 [↗](#) B and E; De Storme et al., 2016 [↗](#)). Combining CENH3 patterns with the histone H3.3 marker showed that somatic cortical cells giving rise to indeterminate Medicago root nodules are mainly tetraploid, which is consistent with previous observations (Torrey & Barrios, 1969 [↗](#)). Furthermore, our imaging data indicate that dividing endopolyploid cells are homogeneously maintained in the nodule tissue (Figure 5 [↗](#)–figure supplement 2B–C). This suggests that tetraploidization is an important feature for nodule fitness and infectability, as it may enable processes such as the required integration of many contextual stimuli (e.g., bacterial signals; Russo & Genre, 2021 [↗](#)), the coordination of controlled organogenesis and a greater cellular adaptability to hypoxic

environments as also suggested for human tumour cells (Yant & Bomblies, 2015 [↗](#)). Furthermore, our findings point towards tetraploidization of cortical cells being a hallmark for nodule identity as pericycle cells in nodule or lateral root primordia remain strictly diploid (**Figure 5E** [↗](#) and **Figure 5** [↗](#)–figure supplement 2C). Interestingly, when dissected and cultured in suspension, cortical cells from the susceptible zone of pea roots entered cell divisions as tetraploid cells only when cytokinin (kinetin) was applied in addition to auxin (2,4-D; Libbenga & Torrey, 1973 [↗](#)). This is particularly interesting since a specific cortical response to cytokinin, an auxin/cytokinin interplay and subsequent cellular proliferation in the root cortex appear to be a legume innovation (Guan et al., 2013 [↗](#); Gauthier-Coles et al., 2019).

NF-YA1, possible intermediary between intracellular infection and cell-cycle control

While this and other reports have shed light onto cell-cycle patterns during rhizobial infections in roots and nodules, genetic factors coordinating cell-cycle progression and intracellular infections remained enigmatic. Using the heterologous CDEL system (Xu et al., 2020 [↗](#)) has enabled us to test the impact of at least one of the potential candidates, the symbiotic transcription factor NF-YA1, on cell cycle-dependent activities in a de-differentiating cellular population. Expression of NF-YA1 in this system did not prevent cells from passing through S-phase (**Figure 7D** [↗](#)). But, while the activated cells (as indicated by the induction of the KNOLLE reporter) still engage in mitosis and eventually divide (**Figure 7** [↗](#)–figure supplement 1B), significantly fewer transformed cells promoted the expression of this G2/M marker at our experimental timepoints (**Figure 7E** [↗](#) and **Figure 7** [↗](#)–figure supplement 1C). These results strongly suggest that, in the CDEL system, NF-YA1 negatively regulates the entry into mitosis despite the presence of CYCD3;1. The latter has been shown to promote mitotic cycles and to restrain differentiation and endoreduplication when accumulated in shoot tissues (Schnittger et al., 2002 [↗](#); Dewitte et al., 2003 [↗](#)). Our observations are also consistent with findings in mammalian cells, where the NF-Y trimer mediates the repression of key mitotic genes in complex with the WT p53 protein (Imbriano et al., 2005 [↗](#)) during a G2 arrest induced by genotoxic stress (Manni et al., 2001 [↗](#)). A link to DNA damage control, potentially arising when endoreduplicated pavement cells are pushed back into a chromosome segregation cycle, seems likely in the CDEL system. However, whether NF-YA1 similarly monitors cell division activity in polyploid nodule primordium cells (**Figure 5E** [↗](#)) and controls the G2/M transition along the infection thread trajectory and in intracellularly colonized nodules (**Figure 6B** [↗](#)) are aspects that need to be addressed independently.

In summary, we propose that a local control of cell-cycle activity and a cell-cycle exit decision along the infection thread trajectory provide a window of cellular competence for cortical infections to progress down to the nodule primordium niche in *Medicago*. We demonstrated that the cell division and differentiation potentials are controlled in a spatially confined cell-file in the root cortex. We also showed that post-replicative, tetraploid cells reaching an 8C DNA content are optimal for intracellular infection. Consequently, it is tempting to hypothesize that exactly these cells comprise a highly specific transcriptional and regulatory landscape that ultimately defines the molecular susceptibility state for intracellular infection competence.

Materials and methods

Plant materials and growth conditions

The *Medicago truncatula* (*Medicago*) wild-type (WT) ecotype A17, cultivar Jemalong was used in this study. The *daphne-like* (FN8113) and *nf-ya1-1* mutant lines have been described previously (Liu et al., 2019b [↗](#); Laporte et al., 2014 [↗](#)). Seeds from the *daphne-like* mutant were kindly provided by Rene Geurts (Laboratory of Molecular Biology, Wageningen University, The Netherlands).

Seeds were scarified by covering them with sulfuric acid (H_2SO_4) 96% for 8 minutes and washed 6 times with sterile tap water. After surface sterilization with a bleach solution (1.2% sodium hypochlorite, NaClO and 0.1% sodium dodecyl sulfate, SDS) for 1 min, the 6 washing steps were repeated and seeds were sown on 1% (w/v) agar plates. Seeds were then stratified at 4 °C for 5 days in the dark before being germinated at 22 °C in the dark for 16 h.

Hairy root transformation

Composite Medicago plants were generated with transgenic hairy roots as previously described (Boisson-Dernier et al., 2001 [\[1\]](#)). *Agrobacterium rhizogenes* ARquaI cells carrying the desired binary vector were cultured in liquid LB medium containing appropriate antibiotics at 28 °C, shaking overnight. Three hundred μl of the culture (OD 600 nm: 0.5-0.7) were plated on solid LB medium with corresponding antibiotics and incubated at 28 °C for 2 days. Plants were prepared by removing the seed coat from germinated seedlings and about 5 mm of the root tip region was removed before the cut site was dipped into the solid *A. rhizogenes* culture. Up to 25 inoculated seedlings were placed on solid Fahræus medium supplemented with 0.5 mM NH_4NO_3 . Transformed seedlings were then grown vertically in a climate chamber at 21° C for 3 days in the dark. After another 4 days at 21 °C (16 h light/8 h dark, 35 $\mu\text{mol m}^{-2} \text{s}^{-1}$ light intensity), with only the roots protected from illumination, composite plants were transferred to fresh solid Fahræus plates (0.5 mM NH_4NO_3) and grown for 10 days at 22 °C following the same photoperiod. Transgenic roots were visualized based on the fluorescence emitted by nuclear markers (*M.t.* pH3.3::H3.3-mCherry; *A.t.* pH3.1::H3.1-mCherry; pLjUbi::NLS-2xmCherry) using an AXIO Zoom.V16 stereomicroscope (Zeiss). Un-transformed roots were removed and the composite plants were transferred in pots (7×7×8 cm) containing a 1:1 (v/v) mixture of quartz sand (0.1–0.5 mm grain size, Sakret) and vermiculite (0–3 mm, Ökohum) equilibrated with 25 ml of a ¼ Hoagland solution (0.1 mM KNH_3). After 4 days of acclimation (16/8 h light/dark photoperiod, 24 °C temperature, 60% humidity and 70-90 $\mu\text{mol m}^{-2} \text{s}^{-1}$ light intensity), plants were subjected to rhizobial inoculation.

For this, *Sinorhizobium meliloti* (Sm2011) strains producing mCherry or GFP were grown on solid TY medium containing 10 $\mu\text{g/ml}$ tetracycline for 2 days at 28 °C. A single colony was used as an inoculum and bacteria were grown in liquid TY medium containing the antibiotic at 28 °C, shaking for 2 days. Cells were centrifuged for 8 min at 2205 g, washed once with liquid Fahræus medium and finally resuspended into the same medium to reach an OD 600 nm of 0.0015. Five ml of this bacterial suspension were applied to each composite plant. Pots were watered twice a week with tap water and fertilized with a ¼ Hoagland solution (0.1mM KNO_3) once a week. Transgenic roots were harvested at different timepoints (8 to 14 days) for tissue fixation, clearing and deep tissue imaging by confocal microscopy.

Construct design

For expressing the canonical (*H3.1*) and replacement (*H3.3*) histone genes of Medicago, genomic fragments containing the promoters (*H3.1*: 2623 bp; *H3.1* (2): 2580 bp; *H3.3*: 2739 bp) and open reading frames (ORF) except the termination codon were used. The pH4::GUS reporter used in quantitative fluorometric assays comprised a 2080 bp fragment of the promoter. Expression of the centromeric histone (*CENH3*) gene (coding sequence) was driven by a tentative promoter comprising 1000 bp upstream of the start codon. Transcriptional KNOLLE reporters were driven by a 2325 bp-long fragment of the *KNOLLE* promoter. The core pentamer of all three mitosis-specific activator (MSA) cis-elements (AACGG) within the *KNOLLE* promoter was mutated by changing two bases (ATTGG) according to Ito et al., 2001 [\[2\]](#). The *KNOLLE* genomic sequence was used to build the translational reporter (pKNOLLE::eGFP-KNOLLE). For Medicago NF-YA1, the genomic sequence was used and the NF-YA1^{mutDBD} variant was created by replacing the conserved RGC amino acid residues (positions 213-215) by an AAA triplet according to Laloum et al., 2014 [\[3\]](#). For the repressor-type MYB3R transcription factor, the coding sequence (CDS) was used. All Medicago sequence information was obtained from the Mt4.0v1 genome through Phytozome.

The *Arabidopsis thaliana* (Arabidopsis) pCYCB1;2::N-CYCB1;2-NLS-3xVenus reporter construct was designed according to Trolet et al., 2019 [\[1\]](#) and included a genomic fragment (AT5G06150) of 1929 bp, comprising 1090 bp of promoter sequence, the 5' untranslated region and 839 bp from the ORF including the first intron. Addition of 2 bases (GG for Gly) at the 3' end of this genomic fragment enabled the production of the N-terminal CYCB1;2 domain (including the destruction box [D-box] motif) fused in-frame to a triple-Venus nuclear reporter (Binder et al., 2014 [\[2\]](#); Nadzieja et al., 2019 [\[3\]](#)). The CYCB1;1 N-terminal (D-box) domain was produced from a genomic fragment (AT4G37490) of 713 bp including the first 2 introns (nucleotides [nts] 41-308 and 488-584) and 129 bp of the third exon (nts 585-713). This module was fused in-frame to a triple-Venus nuclear FP to generate a destabilized version of the fluorescent reporter. The CYCD3;1 (AT4G34160) CDS was placed under control of the cauliflower mosaic virus (CaMV) 35S promoter (426 bp) and fused in frame to a 3xHA module to build the p35S::CYCD3;1-HA construct. Arabidopsis sequence data can be found on Phytozome (*Arabidopsis thaliana* TAIR10 annotation version). To produce the C-terminally truncated, hyperactive version of the *Nicotiana tabacum* MybA2 transcriptional activator (pLjUbi::eGFP-NtMybA2Δ630), the CDS (GenBank accession: AB056123.1) coding for the first 630 amino acid residues was used (Araki et al., 2004 [\[4\]](#)). The GoldenBraid PlacCI transformation plasmid was kindly provided by Bénédicte Desvoyes and Crisanto Gutierrez (Centro de Biología Molecular Severo Ochoa, CSIC-UAM, Madrid, Spain).

Construct production using the Golden Gate cloning system

The Golden Gate modular cloning (MoClo) system was used to generate binary plasmids for transgene expression in plant cells (Weber et al., 2011 [\[5\]](#)). Promoter, genomic and coding sequences were synthesized and cloned into pMS-RQ (GeneArt, Thermo Fisher Scientific, Regensburg, Germany). Level 0 (L0) functional modules used in this study are held for distribution in the ENSA project collection (<https://www.ensa.ac.uk/> [\[6\]](#)). Details of the L1 expression cassettes and L2 binary vector composition, including used promoters, genetically encoded fluorescent proteins (stable or short-lived) or epitope tags and the accession numbers for requesting constructs, are provided in a supplementary file (Table S1). One-pot reactions using the Type IIS restriction enzymes BsaI-HFv2 (R3733S) and BpiI / BbsI-HF (R3539S) and the T4 DNA ligase (M0202S), all from New England BioLabs (Frankfurt am Main, Germany), were assembled according to Binder et al., 2014 [\[2\]](#). Cut-ligation reactions were cycled 50 times between 37 °C for 2 min and 25 °C for 5 min, followed by 5 min at 50 °C and 5 min at 80 °C. Five µl of the reaction were transformed into *Escherichia coli* Stbl3 (for L1 and L2 plasmids carrying histone H3.1 or H3.3 expression cassettes) or TOP10.

Root colonization by *Rhizophagus irregularis*

Following excision of un-transformed roots, composite Medicago plants were transferred to chive (*Allium schoenoprasum*) nurse pot (9 x 9 cm) systems containing *R. irregularis* (DAOM 197198, sterile spore inoculum from Agronutrition, France) set up according to Demchenko et al., 2004 (4 plants x pot). Each pot was covered with a transparent plastic bag (20 x 30 cm) for two days to keep plants in moisturized conditions. Plants were grown in a controlled environment chamber at 24 °C with a 16/8 hour light/dark photoperiod, with each pot being watered every second or third day with 50 ml deionized water and fertilized once per week with 50 ml ¼ Hoagland medium. After 7 to 15 days of co-culturing, composite plants were harvested and excised transgenic roots were subjected to fixation, clearing and Calcofluor White staining prior to imaging.

Clearing and staining of the root material

The presence of all expected fluorescent reporters was verified in the meristematic region using an AXIO Zoom.V16 stereomicroscope (Zeiss) before further analyzing the transgenic roots. Fixation, clearing and cell-wall counter-staining procedures were performed according to Ursache et al., 2018 [\[7\]](#). Briefly, transformed roots were isolated and fixed in a freshly prepared 4% paraformaldehyde (PFA) solution in 1x phosphate-buffered saline (PBS), infiltrated under vacuum

for 15 min and then gently agitated for 60-120 min at room temperature in the dark. After 2 washes in 1x PBS, roots were transferred to the clearing (ClearSee) solution and incubated 7 to 10 days with gentle shaking, changing the ClearSee solution every 2 days. Prior to imaging, roots were stained with 0.1% Calcofluor White in ClearSee for 1 h, before being rinsed once and washed in the clearing solution for 30 min. Calcofluor White was imaged using a 405 nm laser diode excitation source and a 425– 475 nm detection window.

Nodule sections

To assess the localization pattern of H3.1 and H3.3 histone variants across different nodule zones, composite plants were generated by hairy root transformation as described above. Nodules were harvested 35 to 40 days after inoculation in open pots and fixed in 4% (w/v) PFA in 1x PBS, under vacuum treatment for 15 min and then gently agitated for 45 min at room temperature. The fixed samples were embedded in 6% (w/v) low melting temperature agarose (Biozym) and longitudinal sections (70 μ m thickness) were obtained using a VT1000S vibratome microtome (Leica Biosystems, Mannheim, Germany).

Fluorescent reporter imaging by Confocal Laser-Scanning Microscopy (CLSM)

Imaging of plant material (*i.e.*, nodule semi-thin sections and cleared root samples) was performed using a TCS SP8 confocal microscope (Leica Microsystems, Mannheim, Germany) controlled by the LAS X v3 software. Images were acquired with a 20x/0.75 or a 40x/1.10 (HC PL APO CS2) water immersion objective. Genetically encoded FPs were excited with an argon laser or a White Light Laser (WLL) source. Emitted photons were collected using hybrid detectors (HyDs). eCFP was excited with an argon laser line at 458 nm and the emission was detected at 470-510 nm. All other FPs were excited with WLL excitation (ex) lines and emissions (em) were detected according to the following settings: eGFP = 488 nm (ex) / 500-550 nm (em); YFP = 515 nm (ex) / 525-586 nm (em); mCitrine and Venus = 515 nm (ex) / 525-550 (em); mCherry: 561 or 587 nm (ex) / 600-650 or 620-650 nm (em).

Image analysis and quantification of nuclear fluorescent signals

All images were processed and analyzed using the Fiji/ImageJ open-source software (Schindelin *et al.*, 2012 [\[1\]](#)). Image files were used to create figure panels through the FigureJ plugin (Mutterer & Zinck, 2013 [\[2\]](#)). If not indicated differently, 2D maximum intensity projections of z-stacks are presented in the figures. All quantitative measurements were performed on single focal planes and single channels from un-processed z-stacks, acquired from fully transformed Medicago roots. The Fire lookup table (LUT) was applied to facilitate the visualization of low-intensity pixels. Nuclear areas were measured at the equatorial plane using the freehand selection tool, on channels displaying the signal from chromatin-associated proteins (*i.e.*, H3.1 or H3.3-FP fusions). H3.1-eGFP levels were quantified in the nucleus of infected cells (IC) and of directly adjacent, neighbouring cells (NC) present on the same z-stack in the same cortical layer. For this, an outline was manually drawn around each nucleus at the equatorial plane and area, integrated density and mean gray values were measured. Three oval selections were drawn adjacent to all measured nuclei to be used for normalization against background fluorescence. The corrected total nuclear fluorescence (CTNF) = integrated density – (area of selected nucleus $\times$ mean fluorescence of background readings) was calculated according to McCloy *et al.*, 2014 [\[3\]](#) using Excel. The number of mCitrine-CENH3 signals in individual nuclei and mitotic figures was manually determined across image stacks using the multi-point tool. The length of CENH3 fluorescent doublets was measured between the 2 most distant pixels using the straight-line tool.

Transient expression in *Nicotiana benthamiana* leaves

Four to 5-week-old tobacco (*N. benthamiana*) plants were transiently transformed with *A. rhizogenes* (ARqua1) cells carrying the construct of interest. The first, second and third fully expanded true leaves were used for infiltration experiments using needle-less syringes. Agrobacteria were resuspended in a MES/KOH infiltration buffer (10 mM, pH 5.6) supplemented with 10 mM MgCl₂ and 150 μM acetosyringone. Bacterial cultures used for co-transformations were diluted to a final OD 600 nm of 0.1 for each construct. *Agrobacterium tumefaciens* (GV3101) cells producing the p19 silencing suppressor (Silhavy et al., 2002 [DOI](#)), were systematically included at a final OD 600 nm of 0.05. Upon infiltration, plants were further grown for 64 to 72 h before harvesting leaf samples with a biopsy puncher (4 mm diameter) in the distal half of fully infiltrated leaves. Four to 5 leaf discs were either mounted in sterile tap water and imaged with a confocal microscope, or 10 leaf discs were frozen in liquid nitrogen prior to *in vitro* transactivation assays.

Cell division-enabled leaf system (CDELS)

To foster differentiated pavement cells to enter the cell division cycle, non-fluorescently-tagged Arabidopsis CYCD3;1 was expressed under control of the CaMV 35S promoter by Agrobacterium infiltration (Xu et al., 2020 [DOI](#)). The p35S::CYCD3;1-HA construct was co-transformed with binary vectors allowing native expression of cell cycle regulated-reporters (pH4::GUS, pKNOLLE::GUS, pKNOLLE::eGFP-KNOLLE) and constitutive expression of transcription factors (NF-YA1, NF-YA1^{mutDBD}) from Medicago. All plasmids used in the experiments involving the CDEL system are listed in Table S1. Transformed leaves homogenously expressing nuclear transformation markers included on the corresponding binary vectors were selected using an AXIO Imager.M2m ApoTome.2 light microscope (Zeiss). The activity of HA-tagged CYCD3;1 was verified by the presence of at least one mitotic figure with chromosomes labeled by H3.3-mCherry before harvesting.

Immunoblot analyses

To detect proteins transiently produced in the CDEL samples, 3 discs from 64 h post-infiltration leaves were frozen in liquid nitrogen and ground to powder using a Tissue Lyser MM300 (Retsch). The powder was resuspended in 120 μl of 2x loading buffer and heated for 8 min at 95 °C. Equal volumes of samples were loaded and separated by 10% SDS-polyacrylamide gel electrophoresis (PAGE) at constant voltage (80V). Proteins were transferred to a PVDF membrane (0.2 μm pore size) using the Trans-Blot® Turbo™ semi-dry transfer system (7 min mixed MW MIDI program) (Bio-Rad). After blocking in a 2.5% (w/v) milk/TBST solution for 1 h at RT, transferred membranes were hybridized overnight at 4 °C with the following primary antibodies: anti-GFP (Takara Bio Cat# 632381, RRID:AB_2313808) (1/5000); anti-DsRed (Takara Bio Cat# 632496, RRID:AB_10013483) (1/5000); peroxidase-conjugated anti-HA (Roche Cat# 12013819001, RRID:AB_390917) (1/2000); anti-FLAG M2 (Sigma-Aldrich Cat# F3165, RRID:AB_259529) (1/2000). Membranes were washed and incubated 1 h at RT with peroxidase-conjugated secondary antibodies (anti-mouse, Sigma-Aldrich Cat# A4416, RRID:AB_258167; anti-rabbit, Sigma-Aldrich Cat# A6154, RRID:AB_258284) (1/2000). Immunoblotted proteins were detected using the Clarity Western ECL Substrate (Bio-Rad). Digital images were acquired using an ECL ChemoCam Imager (Intas Science imaging) with exposures stopped before saturation.

Transactivation and quantitative fluorometric GUS assays

N. benthamiana leaf samples used for GUS fluorometric assays were harvested 72 h post-infiltration. The same number of plasmids was used for all co-transformation experiments. A construct driving the accumulation of a nuclear FP (proH3.1::NLS:3xVenus) was included in the absence of exogenous transcription factors. Ten leaf discs per transformed leaf were frozen in liquid nitrogen inside a Safe Seal Micro tube containing a stainless-steel bead (5 mm diameter). Samples were ground twice for 30 s (28/s frequency) using a Tissue Lyser MM300 (Retsch) and the

powder was resuspended in 400 µl of cold GUS extraction buffer (50 mM NaHPO₄, pH 7.0, 1 mM EDTA, 10 mM DTT, 0.05% [v/v] Triton X-100 and 1x Protease Inhibitor Cocktail [Roche]). Crude leaf extracts were incubated for 2 h at 4 °C on a turning wheel and centrifuged at 16,900 g for 5 min at 4 °C. Ten µl of the supernatant were used to determine protein concentrations by the dye-binding Bradford assay (Bio-Rad). Twenty-five µg of total proteins in a final volume of 100 µl were pipetted into a white 96 well plate (Greiner Bio-One) before the addition of 200 µl of GUS reaction buffer (50 mM NaHPO₄, pH 7.0, 1 mM EDTA, 1% [v/v] Triton X-100 and 0.5 mM 4-Methylumbelliferyl-β-D-glucuronide [4-MUG] [Carl Roth]). After 10 min of incubation at 37 °C in the dark, the fluorescence emitted by GUS-mediated hydrolysis of 4-MUG into 4-Methylumbelliferone (4-MU) was measured using a microplate reader (POLARstar Omega, BMG LABTECH). Fluorescence was recorded at 360 nm excitation and 460 nm emission filter (gain 500, 20 flashes per well) every 5 min over a time period of 130–180 min. Samples were run in technical triplicates and blank measurements (100 µl extraction buffer + 200 µl reaction buffer) were performed to correct for any nonenzymatic hydrolysis of 4-MUG. Corrected GUS activities (4-MU fluorescence units [FU] min⁻¹) were calculated as: $A_{GUS} = \Delta F / \Delta t = (F1 - F0) / \Delta t$ where ΔF is the difference in fluorescence intensities measured in the linear range of the fluorescence curves over a 15 min period of time (Δt). Values were normalized to 1 µg of protein content by using the protein concentration determined via Bradford assay. Two to 5 independent experiments were performed with at least 3 leaves from 2 different plants analyzed.

Data visualization and statistical analysis

Alignment of multiple protein sequences was performed using Jalview (version 2.11.2.6) (Procter et al., 2021 [DOI](#)). Statistical analyses and generation of figures were performed using the GraphPad Prism software (version 9.5.1) (GraphPad Software Inc.). Normal distribution of data was tested using the Shapiro-Wilk and Kolmogorov-Smirnov normality tests. A Brown-Forsythe and Welch ANOVA test followed by Dunnett's multiple T3 comparisons or an unpaired t-test with Welch's correction were applied as parametric tests. Kruskal-Wallis followed by Dunn's multiple comparisons or Mann-Whitney tests were applied as non-parametric tests. Boxes in the box-and-whiskers plots extend from the 25th to 75th percentiles and whiskers range from the smallest to the largest values. Middle horizontal lines are plotted at the median. Sample size n, statistical tests and significance levels are provided in the figure legends. Source data and results of the statistical analyses are provided as supplement files associated to the corresponding figure.

M. truncatula gene identifiers

Accession numbers for the Medicago genes used in this study are provided from the following genome portals, Phytozome (Mt4.0v1 version; <https://phytozome-next.jgi.doe.gov/> [DOI](#)) and / Medicago truncatula A17 r5.0 (A17 r5.1.9 version; <https://medicago.toulouse.inra.fr/MtrunA17r5.0-ANR/> [DOI](#)), respectively: *H3.1* (1) (Medtr8g092720 / MtrunA17_Chr8g0383781); *H3.1* (2) (Medtr8g103245 / MtrunA17_Chr8g0390361); *H3.3* (Medtr4g097175 / MtrunA17_Chr4g0054161); *H4* (Medtr7g099610 / MtrunA17_Chr7g0262791); *KNOLLE* (Medtr5g012010 / MtrunA17_Chr5g0398891); *CENH3* (Medtr8g027840 / MtrunA17_Chr8g0347231); repressor-type *MYB3R* (Medtr7g061330 / MtrunA17_Chr7g0238631); *NF-YA1* (Medtr1g056530 / MtrunA17_Chr1g0177091).

Data availability statement

All data generated and analyzed during this study are included in the manuscript and supporting files. All sequences have been obtained from publicly available repositories and indicated in the Materials and Methods section.

Materials availability statement

All material generated within this study can be obtained from the corresponding authors upon request. Plasmids obtained from other laboratories as indicated in the respective sections need to be requested from the original producer.

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Conceptualization: MB, TL, TO; Investigation: MB, BL, NZ, CS, SE, PK, JS, FAD; Writing – Original Draft: MB, TO; Writing – Review & Editing: MB, BL, NZ, CS, SE, PK, JS, FAD, TL, TO; Supervision: MB, TL, TO; Project administration: TO; Funding Acquisition: TL, TO

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Reviewer #1 (Public Review):

Many studies reported findings implying that rhizobial infection is associated with cell cycle re-entry and progression, however, our understanding has been fragmented. This study provides exciting new insights as it represents a comprehensive description of the cell cycle progression during early stages of nodulation using fluorescence markers.

To briefly summarize, the authors first monitor H3.1 / H3.3 replacement to distinguish between replicating (S phase) and non-replicating cells to show that *M. truncatula* cortex cells along the bacterial infection thread are non-replicating (while neighbors enter the S phase). Nuclear size measurements revealed that these non-replicative cells are in the post-replicative stage (G2) rather than in the pre-replicative G1 phase, which the authors confirm with the Plant Cell Cycle Indicator (PlaCCI) fluorescent marker to track cell cycle progression in more detail. Cortex cells in the trajectory of the infection thread did not accumulate the late G2 marker of the PlaCCI nor the G2/M marker KNOLLE, indicating that these cells indeed remain in G2. Because nuclear size measurements indicated that infected cells are polyploid, the authors used the centromere histone marker CENH3 to determine chromosome number. They find that cortex cells giving rise to the nodule primordium are endomitotic and tetraploid, probably because their cell cycle is halted at centromere separation. Although not a focus of this manuscript, the authors also use their fluorescent tools to track cell cycle progression during arbuscular mycorrhiza symbiosis. They confirm that infected cells transition from a replicating to a non-replicating state (H3.1 to H3.3) with progressing development of the arbuscules. In addition, the CENH3 marker confirms previous findings that cortex cells infected by fungi are endocycling (i.e., DNA synthesis without segregation of replicated parts). This represents an important confirmation of previous findings and contrasts with the situation during nodulation symbiosis, where chromosomes separate after replication.

In my view, the part about NF-YA1 is less strong - although I realize this is a compelling candidate to be a regulator of cell cycle progression, the experimental approaches used to address this question falls a bit short, in particular, compared to the very detailed approaches shown in the rest of the manuscript. The authors show that the transcription factor NF-YA1 regulates cell division in tobacco leaves; however, there is no experimental validation in the experimental system (nodules). All conclusions are based on a heterologous cell division system in tobacco leaves. The authors state that NF-YA1 has a nodule-specific role as a regulator of cell differentiation. I am concerned the tobacco system may not allow for adequate testing of this hypothesis. With the fluorescent tools the authors have at hand (in particular tools to detect G2/M transition, which the authors suggest is regulated by NF-YA1), it would be interesting to test what happens to cell division if NF-YA1 is over-expressed in Medicago roots?

Based on NF-YA1 expression data published previously and their results in tobacco epidermal cells, the authors hypothesize that NF-YA regulates the mitotic entry of nodule primordial cells. Given that much of the manuscript deals with earlier stages of the infection, I wonder if NF-YA1 could also have a role in regulating mitotic entry in cells adjacent to the infection thread?

In general, all microscopy images are of very high quality and support the authors' conclusions. While individually each set of fluorescent markers has its limitations, combined they constitute a powerful tool to track various stages of cell cycle progression in individual root cells during symbiosis. Overall, this is a very strong manuscript that comprehensively elucidates root cell cycle changes during microbial infection.

- <https://doi.org/10.7554/eLife.88588.1.sa1>

Reviewer #2 (Public Review):

Cell cycle control during nitrogen-fixing symbiosis is an important topic, but our understanding of the process is poor and lacks resolution, as the nodule is a complex organ with many cell types that undergo profound changes. The authors aim to define the cell cycle state of individual plant cells in the emerging nodule primordium, as a transcellular infection thread passes through the meristem to reach cells deep in the incipient nodule and releases bacteria to form symbiosomes. The authors used a number of cell cycle reporters, such as different Histone 3 variants and cyclins, to follow cell cycle progress in exquisite detail. They showed that the host cells in the path of an infection thread exhibit a cell fate distinct from their immediate neighbors: after entering the S phase similar to their neighbors, these cells exit the cell cycle and enter a special differentiated state. This is likely an important shift that allows the proper passage of the infection thread. Although definitive proof needs more investigation, they showed that a pioneering transcription factor, NF-YA1, likely represses these endoreduplicated cells from completing the cell cycle.

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Author Response:

We would like to thank you very much for handling and reviewing our manuscript so carefully and to be so positive about our work. We are indeed grateful about these very concise and constructive reviews as well as about the Editorial Assessment. We basically agree with all reviewers' comments. Besides addressing all formal suggestions, we also decided to do some more experiments.

The main concern, the role of the transcription factor NF-YA1 during rhizobial infections, is indeed an absolut valid one. While the CDEL system has its beauties it certainly has its limitations as well. Thus, we will try to assess the role of NF-YA1 during symbiotic infections in *Medicago* more specifically. We will place NF-YA1 expression under the control of infection-specific promoters to limit pleiotropic effects of ectopic over-expression and assess rhizobial infections as well as cell cycle patterns in tranformed hairy roots producing the H3.1/H3.3 marker. Infection-inducible promoters will also be used to drive the ectopic expression of CYCD3;1 on the cortical infection thread trajectory to locally increase mitotic cycles, in order to test the functional importance of cell cycle exit on cortical infections.

We hope that we will be able to conclude more firmly on NF-YA1 function prior to locking the version of record and to deliver these experiments in a time frame of about 4-6 months, which is the minimum time we need for cloning the respective constructs, doing all hairy root transformations in sufficient numbers and quantitative microscopy.